

RISK AND DECISION BASELINE

Trade-Off Analysis Matrices

Natural-Language Treatment Planning System (NL-TPS)

Source baseline: NL-TPS controlled documentation suite and Risk Register v0.1

Version: 0.1 - Preliminary engineering baseline

Date: 18 August 2026

Status: Discussion Draft - Not for clinical use

Coverage: 1,948 entity-level matrices: 1,776 requirements and 172 component responsibilities

STATUS AND SAFETY NOTICE

These matrices are preliminary systems-engineering decision screens. They do not waive requirements, approve architecture, accept clinical risk, select a vendor or model, establish regulatory conclusions, or authorize patient use. Formal decisions require approved alternatives, evidence, uncertainty and sensitivity analysis, and accountable governance.

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Document control

Field	Value
Document owner	NL-TPS Program - proposed; final owner requires governance approval
Source baseline	NL-TPS controlled documentation suite and Risk and Mitigation Register, Version 0.1
Coverage	1,948 entity-level matrices: 1,776 requirements and 172 component responsibilities
Risk status	Preliminary failure-mode and control screen; not an approved FMEA, hazard analysis, residual-risk decision, or clinical risk acceptance
Change authority	Clinical governance, risk management, quality system, and configuration control
Supersession	None; source requirements and component documents remain authoritative

Executive summary

This document provides one traceable trade-off matrix row for each of the 1,948 unique requirements and components covered by the risk register. Every row identifies the decision focus, applicable alternative set, six-criterion score vector, weighted result for three alternatives, preferred preliminary disposition, and the condition that can change the decision.

Requirements remain mandatory. A trade study selects a compliant realization; it cannot vote away a shall-clause, human-authority boundary, independent QA obligation, commissioned-TPS role, DICOM/DICOM-RT validation, risk control, or release condition. A noncompliant alternative is infeasible regardless of score.

PRELIMINARY DECISION STATUS

Scores are structured prompts for engineering and clinical discussion, not validated measurements or automatic decisions. Project-specific evidence shall replace the preliminary scores. The decision record shall document alternatives, criteria, weights, assumptions, uncertainty, sensitivity, conflicts, risk effects, recommendation, decision authority, and implementation conditions.

1 Decision-analysis method

The matrices use weighted multi-criteria decision analysis. Each criterion is scored from 1 (least favorable) to 5 (most favorable). Weighted total is the sum of score multiplied by criterion weight, divided by 100. Ties or differences below an institution-defined meaningful threshold require sensitivity analysis rather than automatic selection.

Code	Criterion	Weight	Question
S	Patient and use safety	30	How well does the alternative prevent, detect, contain, and recover from patient, user, authority, or clinical harm?
U	Clinical and workflow utility	15	Does it support intended users, clinical goals, timely review, professional authority, and usable recovery?
I	Data and interoperability fidelity	15	Does it preserve identity, meaning, units, geometry, versions, references, provenance, and external-TPS semantics?
A	Assurance and auditability	15	Can it be specified, independently tested, monitored, reconstructed, commissioned, and revalidated?
L	Lifecycle and resilience	15	Can it be changed, supported, secured, recovered, scaled, and sustained without unacceptable common-cause or supplier risk?
C	Cost and schedule feasibility	10	What staffing, integration, validation, infrastructure, licensing, support, and delivery burden does it impose?

2 Alternative sets

Set	Alternative A	Alternative B	Alternative C
R	Primarily procedural or shared-service compliant realization with checklist, explicit human control, and retained records.	Bounded deterministic workflow or service with versioned policy, human gate, audit, monitoring, and verification.	Independent or redundant validator/control with isolation, diverse check where justified, automated reconciliation, and containment.
I	Endpoint-local contract and validation with controlled manual reconciliation and route-specific commissioning.	Shared versioned contract, trusted gateway, deterministic validation, atomic transaction control, explicit outcome, and reconciliation.	Independent producer and consumer validation plus shadow or dual comparison, automated quarantine and reconciliation, and enhanced continuity.
C	Consolidate the logical responsibility into an adjacent service or team under an explicit internal contract and evidence boundary.	Maintain a distinct logical component responsibility with stable contract, owner, configuration, monitoring, failure behavior, and independent test boundary.	Deploy isolated or redundant instances/control planes with independent validation, recovery, and enhanced separation of duties.
O	Controlled manual procedure and records with trained roles, checklist, escalation, and periodic review.	Governed workflow tooling with accountable approvals, controlled automation, audit, monitoring, and exercise evidence.	Independent automated enforcement plus secondary oversight, resilience, continuous monitoring, and tested continuity.

All alternatives are conditional on full compliance. Alternative A is infeasible when a deterministic, automated, independent, real-time, or commissioned control is explicitly required. Alternative C is not inherently safer if it adds common-cause software, latency, alert burden, inconsistent states, new attack surface, or operational complexity.

3 Standards and evidence basis

Reference	Use in this baseline	Controlled source
AAPM TG-100 / Report 283	Process mapping, failure modes, effects, causes, controls, FMEA, fault-tree analysis, and risk-informed radiotherapy QM.	AAPM official report page
ISO 14971:2019	Lifecycle risk management, risk analysis and evaluation, risk controls, residual risk, production and post-production information.	ISO official standard page
FDA benefit-risk guidance	Transparent consideration of benefit, risk, uncertainty, controls, evidence, and total-product-lifecycle information.	FDA official guidance page
NASA Systems Engineering Handbook	Decision analysis and trade studies: define alternatives and criteria, document assumptions, uncertainty, sensitivity, recommendation, impacts, and final decision.	NASA official handbook
DICOM PS3 and IHE-RO	Paired conformance, semantic, role, version, security, workflow, and site-specific interoperability risk at external TPS boundaries.	DICOM official source ; IHE-RO official source
Approved clinical evidence	AAPM task-group reports, QUANTEC/HyTEC, peer-reviewed disease-site guidance, prescriptions, and institutional policy remain governed inputs, not universal thresholds in this document.	ConOps EVID and source HLR EVD

4 Coverage and preliminary disposition summary

Entity set	Count	C	B + targeted C	B	Status
System high-level requirements	111	31	79	1	Complete
Detailed sub-requirements	333	47	229	57	Complete
High-level interface requirements	333	50	223	60	Complete
Sub-interface requirements	999	150	669	180	Complete
Functional components	41	11	28	2	Complete
Interface components	57	17	34	6	Complete
F/NF/O category components	74	11	55	8	Complete
TOTAL	1,948	317	1317	314	Complete

5 System high-level requirement trade matrices

5.1 Governance, intended use, and scope (GOV) - 7 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
GOV-001	Interoperability and external TPS: operate as a human-supervised treatment-planning support system and shall not be represented as a...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
GOV-002	Human authority and approval: restrict clinical operation to the institution-approved intended use, patient population, disease sites, modalities, machines...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-003	Human authority and approval: block clinical execution for an unsupported or out-of-scope workflow until that workflow has completed...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-004	Human authority and approval: provide segregated training/ research, commissioning, shadow, clinical-draft, clinical-approved, and degraded/read-only operating modes with mode-specific permissions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-005	Human authority and approval: enforce institution-approved role authority and separation-of-duties rules for radiation oncologists, dosimetrists, medical physicists, therapists...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-006	Human authority and approval: maintain the intended-use statement, approved scope, risk controls, release status, configuration baseline, and applicable...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-007	Human authority and approval: require clinical-governance approval for changes to deployment scope, accepted residual risk, evidence policy, model...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.2 Safety boundaries and human authority (SAF) - 10 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SAF-001	Human authority and approval: prevent any AI or automated service from pre-prescribing treatment, signing or approving a prescription...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-002	Patient and case identity: bind every clinical action to an authenticated user and role and to a confirmed...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-003	Patient and case identity: require confirmation of at least two patient identifiers before a state-changing clinical action is...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-004	Evidence and clinical rules: fail closed when a safety-critical field, permission, dependency, context assertion, evidence rule, or commissioned-use...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-005	Anatomy, registration, and geometry: keep automated contours, objectives, transformations, calculations, and candidate plans in a visibly unsigned draft...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-006	Interoperability and external TPS: provide a structured preview, explicit confirmation, cancellation, and rollback for reversible state-changing actions before...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-007	Human authority and approval: invalidate affected downstream approvals, QA evidence, and transfer status whenever an approved dependency changes	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-008	Human authority and approval: implement role authorization, state transitions, context assertions, hard stops, approval invalidation, and export policy...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SAF-009	Prescription, plan, and dose: preserve institutionally required independent review, dose or MU verification, data-transfer verification, patient-specific QA, and...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-010	Evidence and clinical rules: identify degraded operation and failed dependencies explicitly and shall not use an unannounced fallback...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.3 Natural-language and voice interaction (NLI) - 8 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NLI-001	Human authority and approval: accept typed text and user-initiated push-to-talk speech while prohibiting background listening and voice-only clinical...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
NLI-002	Natural language and intent: compile each actionable natural-language request into a versioned, inspectable, typed intent before executing a...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
NLI-003	Human authority and approval: identify the clinical objects, requested action, parameters, units, priorities, evidence references, preconditions, expected outputs...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-004	Evidence and clinical rules: distinguish verified patient facts, signed clinical intent, user preferences, evidence-derived rules, system assumptions, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NLI-005	Prescription, plan, and dose: use deterministic validation for safety-critical numbers, units, dose basis, fractionation, negation, laterality, structure identity...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
NLI-006	Natural language and intent: display a structured read-back of what will change, the affected patient and object, assumptions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-007	Prescription, plan, and dose: require explicit confirmation of speech transcripts containing safety-critical numbers, units, negation, laterality, prescription elements...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
NLI-008	Natural language and intent: record the original user input or confirmed transcript, compiled intent, clarifications, assumptions, preview, confirmations...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.4 Evidence and literature governance (EVD) - 8 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
EVD-001	Human authority and approval: use only an institution-approved, closed evidence service for patient-specific clinical rules and shall not...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-002	Human authority and approval: attach the source title, version or date, exact location, population, applicability, limitations, institutional approval...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
EVD-003	Human authority and approval: apply the approved authority order of patient-specific signed intent, applicable trial protocol, institutional standard...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-004	Prescription, plan, and dose: treat the signed prescription and physician-approved case intent as the highest patient-specific authority and...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
EVD-005	Evidence and clinical rules: block or escalate conflicting, expired, retired, superseded, or in-applicable evidence rather than silently selecting...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-006	Prescription, plan, and dose: represent each computable rule with explicit structure definition, units, comparator, dose quantity and basis...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
EVD-007	Human authority and approval: support controlled evidence proposal, independent clinical review, approval, publication, periodic review, retirement, emergency suspension...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-008	Interoperability and external TPS: provide a no-source or no-applicable-rule response and require human resolution when a verified clinical...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.5 Clinical context, imaging, registration, and contours (CLN) - 9 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
CLN-001	Interoperability and external TPS: map verified case facts to candidate workflows or templates, identify missing material information, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-002	Anatomy, registration, and geometry: acquire and bind the approved image studies, frame of reference, registrations, structure sets, prior-treatment...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
CLN-003	Prescription, plan, and dose: permit rigid or deformable registration and derived dose accumulation only through commissioned methods with...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
CLN-004	Anatomy, registration, and geometry: invoke contouring models only for approved anatomy, imaging conditions, structures, populations, and use cases...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-005	Anatomy, registration, and geometry: display contour-model uncertainty, input anomalies, and out-of-distribution status and shall abstain or route to...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-006	Anatomy, registration, and geometry: validate structure identity, nomenclature mapping, laterality, presence, completeness, topology, image coverage, and geometric plausibility...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-007	Anatomy, registration, and geometry: preserve each contour's input study, frame of reference, model and version, configuration, naming map...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
CLN-008	Interoperability and external TPS: provide slice, surface, difference, and clinically relevant error-review views and shall capture the location...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-009	Anatomy, registration, and geometry: require radiation-oncologist approval for clinical target contours and shall enforce institution-approved approval roles for...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.6 Treatment-plan generation and calculation (PLN) - 12 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
PLN-001	Prescription, plan, and dose: create a versioned structured planning intent from the signed prescription, approved anatomy, patient-specific instructions...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-002	Prescription, plan, and dose: generate or modify candidate plans only through commissioned planning, optimization, robustness, and dose-calculation services...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-003	Prescription, plan, and dose: prevent language, speech, disease-site, or generative models from directly calculating clinical dose or substituting...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-004	Prescription, plan, and dose: distinguish hard constraints, planning goals, optimization objectives, preferences, and reporting-only metrics	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
PLN-005	Interoperability and external TPS: prevent silent relaxation of a hard constraint and shall require an authorized explicit action...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-006	Interoperability and external TPS: generate the number of candidate plans requested by an authorized user when feasible and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-007	Interoperability and external TPS: document the planning strategy, parameter variation, objective tradeoff, beam or modality variation, or other...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-008	Interoperability and external TPS: identify infeasible candidates and candidates dominated by another candidate under the declared evaluation criteria	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-009	Prescription, plan, and dose: attest at runtime to the active machine model, energy or beam type, technique, dose...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-010	Prescription, plan, and dose: represent physical dose, RBE-weighted dose, fractionation, normalization, BED or EQD2 when approved, and proton-specific...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-011	Prescription, plan, and dose: apply and record the institution-approved setup, range, motion, interplay, and other robustness scenarios required...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-012	Interoperability and external TPS: pre-serve candidate-plan identity, parent intent, parameters, calculation outputs, warnings, configuration, evaluation results, and lineage...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
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5.7 Plan review, selection, QA, and transfer (REV) - 9 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
REV-001	Interoperability and external TPS: present candidate plans in a consistent comparison workspace showing absolute values and differences without...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-002	Prescription, plan, and dose: display target coverage, organ-at-risk results, constraint status, spatial dose, hotspots and coldspots, robustness, uncertainty...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
REV-003	Interoperability and external TPS: expose the declared tradeoffs and shall not select or recommend a winning plan solely...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-004	Interoperability and external TPS: record the selected plan, rejected alternatives, review comments, deviations, tradeoff rationale, and role-specific decisions	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-005	Human authority and approval: enforce radiation-oncologist clinical plan approval and qualified-medical-physicist technical review before a candidate can enter...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-006	Prescription, plan, and dose: route the approved candidate through the institutionally required independent dose or MU verification, patient-specific...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
REV-007	Anatomy, registration, and geometry: authorize export only after all required approvals and QA states are current and shall...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-008	Human authority and approval: make approved clinical objects immutable or version-controlled and shall require re-review of every affected...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-009	Interoperability and external TPS: document the independence assumptions, shared inputs, shared dependencies, ownership boundaries, and common-cause failure risks...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.8 Data, interoperability, provenance, and audit (DAT) - 10 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
DAT-001	Prescription, plan, and dose: assign a unique identity, version, lifecycle state, owner, and source linkage to each clinical...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
DAT-002	Anatomy, registration, and geometry: maintain a dependency and provenance graph that traces every derived object to its inputs...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-003	Prescription, plan, and dose: validate and preserve explicit units, coordinate systems, dose quantity and basis, structure definitions, and...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
DAT-004	Anatomy, registration, and geometry: validate DICOM patient and study identities, SOP Instance UIDs, references, frame-of-reference UIDs, required attributes...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
DAT-005	Prescription, plan, and dose: support the approved DICOM objects and transactions required for images, registrations, segmentations or structures...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
DAT-006	Human authority and approval: use approved IHE-RO profiles and FHIR or CodeX radiation-therapy exchanges where implemented and shall...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-007	Interoperability and external TPS: reject incomplete, ambiguous, inconsistent, orphaned, or partially transferred clinical data rather than promoting it...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-008	Human authority and approval: maintain a tamper-evident, time-synchronized audit ledger of commands, context, inputs, versions, evidence, outputs, warnings...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-009	Evidence and clinical rules: use integrity hashes or equivalent controls to detect unauthorized modification of clinical objects, evidence...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-010	Interoperability and external TPS: identify the authoritative system of record and retention, legal-record, export, archival, and reconciliation policy...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.9 AI and inference-model lifecycle (AIM) - 8 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
AIM-001	Human authority and approval: invoke only institution-approved, locked model versions registered for the active task and clinical operating...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-002	AI model and inference: verify that the active case satisfies the model's intended use, input contract, population, imaging...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-003	AI model and inference: implement a defined abstention, rejection, or manual-review response for unsupported, anomalous, incomplete, or out-of-distribution...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-004	Human authority and approval: prohibit production model-weight updates and unapproved behavioral learning from routine patient cases, user edits...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-005	Human authority and approval: maintain a model and dependency registry containing version, intended use, owner, supplier, training and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-006	AI model and inference: use separated development, tuning, internal test, external test, and site-acceptance data with documented overlap...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-007	Human authority and approval: evaluate and display clinically relevant overall and subgroup performance, calibration, failure detection, uncertainty, edit...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-008	Human authority and approval: undergo impact assessment, regression testing, approval, versioning, and rollback planning	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
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5.10 Human factors, usability, and competency (HFE) - 6 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
HFE-001	Patient and case identity: continuously display the active patient, course, study, plan, operating mode, approval state, and unsaved...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
HFE-002	Human authority and approval: visually distinguish verified patient facts, signed intent, approved evidence, model suggestions, user preferences, assumptions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
HFE-003	Human factors and use safety: identify the hazard, affected object, required action, and escalation path and shall not rely...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-004	Interoperability and external TPS: be proportional to clinical risk, and the NL-TPS shall monitor alert burden, overrides, workarounds...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
HFE-005	Human authority and approval: complete formative and summative human-factors validation with representative intended users, critical tasks, environments, interruptions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
HFE-006	Interoperability and external TPS: require role-specific training, demonstrated competency, and current authorization before a user can perform clinical...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.11 Privacy, cybersecurity, and secure development (SEC) - 8 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SEC-001	Cybersecurity and privacy: enforce unique user identity, multi-factor authentication, least privilege, role and context authorization, session controls...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-002	Anatomy, registration, and geometry: protect data and services through approved encryption, network segmentation, managed secrets, secure configuration, endpoint...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-003	Human authority and approval: prevent protected health information, images, prompts, transcripts, or outputs from being sent to an...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-004	Human authority and approval: treat retrieved text, uploaded documents, DICOM metadata, user content, and model output as untrusted...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-005	Human authority and approval: verify the identity, integrity, signature, provenance, and approved status of software, model, evidence, configuration...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-006	AI model and inference: include threat modeling, secure coding, code review, software bill of materials, vulnerability scanning, penetration...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-007	Evidence and clinical rules: centrally record and detect unauthorized access, anomalous export, integrity failure, malicious input, model or...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SEC-008	Evidence and clinical rules: support tested clinical and cybersecurity incident containment, evidence preservation, patient-impact assessment, communication, reporting, recovery...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.12 Clinical operations, monitoring, and resilience (OPS) - 7 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OPS-001	Human authority and approval: monitor predefined safety, model, evidence, workflow, technical, subgroup, reliability, and security signals against approved...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-002	Interoperability and external TPS: use atomic job states and shall prevent a timed-out, failed, canceled, partial, or unreconciled...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-003	Human authority and approval: support rapid suspension of a model, evidence rule, connector, workflow, disease site, or complete...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-004	Human authority and approval: provide an explicit degraded or read-only mode and an institution-approved downtime procedure that prohibits...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-005	Human authority and approval: maintain validated backups and shall perform staged restoration, object and destination reconciliation, approval-state verification...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OPS-006	Evidence and clinical rules: support reporting, triage, investigation, patient-impact assessment, evidence preservation, trend analysis, and corrective and preventive...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-007	Human authority and approval: define, approve, monitor, and periodically review clinical service levels and action thresholds for availability...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

5.13 Verification, commissioning, release, and change control (VAL) - 9 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
VAL-001	Human authority and approval: trace each stakeholder need, hazard, risk control, and high-level requirement to allocated design, implementation...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-002	Human authority and approval: pass site-specific acceptance, commissioning, end-to-end, regression, shadow-mode, and governance release criteria before clinical use	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-003	Prescription, plan, and dose: include locked reference cases and challenge cases for language and voice, evidence computation, segmentation...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
VAL-004	Human authority and approval: exercise normal, boundary, rare but foreseeable, wrong-context, partial-failure, time-out, rollback, post-approval change, and recovery...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
VAL-005	Prescription, plan, and dose: include task-specific technical performance, clinically significant error, dose impact where relevant, edit burden, failure...	R	1/3/2/2/5; 2.15	4/4/4/4/3; 3.90	5/4/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
VAL-006	Interoperability and external TPS: progress through retrospective, shadow, and controlled limited-clinical stages with predefined endpoints, independent adjudication, and...	R	2/3/2/2/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-007	Human authority and approval: prevent progression through each release gate until its predefined safety, quality, subgroup, usability, reliability...	R	2/3/2/2/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-008	Human authority and approval: receive documented change-impact assessment, commissioning, human-factors assessment, regression testing, and monitoring approval	R	2/3/2/2/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-009	Human authority and approval: not enter or remain in clinical service with an unresolved critical anomaly, unacceptable residual...	R	2/3/2/2/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6 Detailed sub-requirement trade matrices

6.1 Governance, intended use, and scope (GOV) - 21 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
GOV-001-01	Human authority and approval: identify AI-produced content as decision support or draft output and shall identify the authorized...	R	2/3/2/2/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
GOV-001-02	Human authority and approval: reject any state transition in which model output, an automated service identity, or an...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-001-03	Human authority and approval: review the intended-use statement, labeling, training content, and release communications for consistency with the...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-002-01	Human authority and approval: maintain a versioned approved-use matrix containing the permitted institution, patient population, disease site, modality...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-002-02	Human authority and approval: compare the active case and requested workflow with every applicable field in the approved-use...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-002-03	Human authority and approval: record the evaluated scope attributes, approved-use matrix version, decision, and denial reason for each...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-003-01	Governance, scope, and change: classify any unlisted scope value or workflow combination as unsupported and shall block state...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
GOV-003-02	Human authority and approval: require linked risk assessment, commissioning, validation, and governance-approval records for each proposed scope expansion	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-003-03	Human authority and approval: prevent activation of a scope expansion until all required records are complete, approved, effective...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
GOV-004-01	Human authority and approval: define, for each operating mode, the permitted actions, data classes, model and evidence versions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-004-02	Interoperability and external TPS: enforce the active mode's permissions and shall display the active mode continuously during user...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-004-03	Human authority and approval: prevent promotion or transfer of objects between operating modes except through an approved, versioned...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-005-01	Human authority and approval: implement an institution-approved role-permission matrix that identifies allowed functions and incompatible responsibilities for each...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-005-02	Human authority and approval: prevent one actor or service identity from satisfying approval steps that the active separation-of-duties...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-005-03	Data integrity, provenance, and audit: perform and retain periodic access, role, exception, and separation-of-duties reviews, including disposition of identified...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
GOV-006-01	Human authority and approval: assign an identifier, owner, version, approval state, effective date, and supersession status to each...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-006-02	Human authority and approval: reference the exact approved versions of all required controlled artifacts and shall not activate...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
GOV-006-03	Human authority and approval: generate a reproducible release-baseline report listing each controlled artifact, approval, effective status, and integrity...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-007-01	Human authority and approval: classify changes to deployment scope, residual risk, evidence policy, model release, monitoring thresholds, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-007-02	Human authority and approval: require the designated clinical-governance approvals before a governance-controlled change can be deployed or made...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
GOV-007-03	Human authority and approval: retain the prior and new values, rationale, impact assessment, approvers, approval times, effective date...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.2 Safety boundaries and human authority (SAF) - 30 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SAF-001-01	Human authority and approval: expose no AI-callable capability that can prescribe treatment, provide a clinical signature, waive QA...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-001-02	Human authority and approval: require an authenticated authorized human acting through the designated clinical workflow and shall reject...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SAF-001-03	AI model and inference: demonstrate that prompts, tool calls, model output, malformed requests, and privilege escalation attempts cannot...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-002-01	Patient and case identity: create an immutable action context containing the authenticated user, role, patient, course, study, frame...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-002-02	Interoperability and external TPS: block a clinical action when any required context element is absent or inconsistent, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-002-03	Data integrity, provenance, and audit: store the complete action context and the identifiers of every object read, changed, created...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SAF-003-01	Patient and case identity: present at least two independently sourced patient identifiers and shall require an active user...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-003-02	Human authority and approval: reject state-changing execution when the identifier confirmation is absent, stale, associated with another patient...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-003-03	Patient and case identity: record identifier-confirmation events and shall include wrong-patient, interruption, and context-switch scenarios in human-factors validation	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-004-01	Human authority and approval: represent each safety-critical precondition with explicit satisfied, unsatisfied, or unknown status and shall permit...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SAF-004-02	Interoperability and external TPS: identify each blocking precondition, the affected action, and the authorized resolution path without offering...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-004-03	Verification, commissioning, and release: record every fail-closed decision and shall verify missing, ambiguous, inconsistent, expired, and unsupported preconditions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-005-01	Anatomy, registration, and geometry: assign every automated contour, objective, transformation, calculation, and candidate plan an unsigned draft or...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-005-02	Human authority and approval: distinguish draft and sandbox objects from reviewed and approved clinical objects using persistent text...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-005-03	Human authority and approval: reject an automated object until all configured human reviews and approvals are current	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-006-01	Human authority and approval: identify the affected objects, current and proposed values, expected consequences, required approvals, cancellation path...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-006-02	Human authority and approval: require a distinct confirmation action after preview, and the confirmation shall expire when the...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-006-03	Human authority and approval: execute as recoverable transactions with tested cancellation and rollback behavior that preserves object identity...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SAF-007-01	Human authority and approval: identify every downstream approval, QA result, transfer status, and derived object affected by a...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-007-02	Human authority and approval: atomically invalidate affected approvals, QA evidence, and export eligibility before the changed dependency becomes...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-007-03	Interoperability and external TPS: present the required re-review set, and the audit ledger shall record the invalidation cause...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-008-01	AI model and inference: be a versioned deterministic component separated from generative-model prompts, weights, sampling behavior, and model-provided...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-008-02	Human authority and approval: implement explicit allowlisted rules for authorization, state transitions, context assertions, hard stops, approval invalidation...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-008-03	Evidence and clinical rules: provide an independent test interface and evidence demonstrating repeatable decisions for equivalent inputs across...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-009-01	Prescription, plan, and dose: identify every required independent review, dose or MU verification, transfer verification, patient-specific QA activity...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SAF-009-02	Human authority and approval: prevent AI output, shared automated evidence, or one incompatible actor from satisfying an independent...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SAF-009-03	Evidence and clinical rules: remain blocked until current results and authorized sign-offs exist for every required independent check...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-010-01	Human authority and approval: assign an explicit normal, degraded, failed, or unavailable state to each safety-relevant dependency and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-010-02	Human authority and approval: announce degraded operation and the safety kernel shall prohibit any fallback that is not...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SAF-010-03	Data integrity, provenance, and audit: record the failed dependency, selected response, affected clinical behavior, user notifications, confirmations, and restoration...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

6.3 Natural-language and voice interaction (NLI) - 24 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NLI-001-01	Natural language and intent: accept typed input and shall activate speech capture only while an authenticated user deliberately...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-001-02	Human authority and approval: be disabled by default, shall terminate when push-to-talk is released or canceled, and shall...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-001-03	Human authority and approval: prohibit voice-only clinical signatures and shall route all approval actions through an authenticated non-voice...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NLI-002-01	Anatomy, registration, and geometry: transform each actionable request into a schema-valid typed-intent object with a unique identity, schema...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-002-02	Natural language and intent: prevent a state-changing action until the typed intent has passed deterministic validation and the...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
NLI-002-03	Verification, commissioning, and release: retain every typed-intent version and its relationship to the original input, clarifications, validation result...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-003-01	Human authority and approval: provide explicit fields for clinical-object identifiers, action, parameters, units, priorities, evidence references, preconditions, expected...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-003-02	Verification, commissioning, and release: reject a required field that is missing, untyped, unresolved, internally inconsistent, or not bound...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-003-03	Interoperability and external TPS: render the same serialized typed-intent content used for deterministic execution	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-004-01	Evidence and clinical rules: carry a source classification of verified fact, signed clinical intent, user preference, evidence-derived rule...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-004-02	Interoperability and external TPS: distinguish source classifications with persistent text labels and a visible legend that does not...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NLI-004-03	AI model and inference: prevent a model suggestion, preference, assumption, or unresolved question from overwriting or being promoted...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-005-01	Prescription, plan, and dose: parse and normalize safety-critical numbers, units, dose basis, fractionation, negation, laterality, structure identity, and...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
NLI-005-02	Natural language and intent: reject or request clarification for ambiguous, incompatible, implausible, or unresolved safety-critical values before intent...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-005-03	Natural language and intent: include accepted, rejected, boundary, negated, unit-conversion, laterality, alias, dictation-error, and adversarial examples for every...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-006-01	Natural language and intent: identify the action, affected patient and objects, current and proposed values, assumptions, evidence, expected...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-006-02	Natural language and intent: require the read-back to be generated from the validated typed intent and shall regenerate...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-006-03	Natural language and intent: be retained in the audit ledger	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-007-01	Prescription, plan, and dose: flag utterances containing safety-critical numbers, units, negation, laterality, prescription elements, or clinical-object names	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NLI-007-02	Natural language and intent: be displayed for active user confirmation or correction, and the safety kernel shall block...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-007-03	Natural language and intent: retain the captured transcript, flagged tokens, user corrections, confirmed transcript, and confirmation identity and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-008-01	Natural language and intent: link the original text or confirmed transcript, typed-intent versions, clarification turns, assumptions, previews, confirmations...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-008-02	Human authority and approval: be append-only, integrity protected, access controlled, and time synchronized according to the approved retention...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NLI-008-03	Natural language and intent: be able to reconstruct the ordered interaction and identify the exact inputs, versions, decisions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.4 Evidence and literature governance (EVD) - 24 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
EVD-001-01	Human authority and approval: use an allowlisted evidence repository and retrieval configuration containing only institution-approved sources, rules, versions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
EVD-001-02	Human authority and approval: block unrestricted internet retrieval and unapproved external connectors for patient-specific clinical-rule generation in clinical...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-001-03	Evidence and clinical rules: record the repository, retrieval configuration, source set, operating mode, result, and any blocked retrieval...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-002-01	Human authority and approval: require source title, version or date, exact location, population, applicability, limitations, approval status, effective...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-002-02	Human authority and approval: reject an evidence-derived rule when any required provenance field is missing, invalid, expired, or...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-002-03	Evidence and clinical rules: make the complete provenance and applicability record available from every displayed or applied evidence-derived...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-003-01	Human authority and approval: encode the approved authority tiers and ordering for signed patient intent, trial protocol, institutional...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-003-02	Human authority and approval: apply the highest applicable authority and shall identify unresolved conflicts rather than combining rules...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-003-03	Human authority and approval: display and record the authority tier, applicability determination, selected rule, excluded rules, and conflict-resolution...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
EVD-004-01	Prescription, plan, and dose: bind the current signed prescription and physician-approved case intent as the highest patient-specific authority...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
EVD-004-02	Human authority and approval: issue a hard stop when a lower-tier rule conflicts with signed patient-specific authority and...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
EVD-004-03	Human factors and use safety: identify the conflicting fields, source versions, affected objects, and required physician resolution	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
EVD-005-01	Evidence and clinical rules: have a controlled lifecycle state, effective interval, supersession link, applicability predicate, and conflict status	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-005-02	Evidence and clinical rules: exclude expired, retired, suspended, superseded, and inapplicable rules and shall block or escalate unresolved...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-005-03	Data integrity, provenance, and audit: retain all candidate rules, exclusion reasons, lifecycle states, conflict results, and the human disposition...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
EVD-006-01	Prescription, plan, and dose: contain a controlled structure definition, units, comparator, dose quantity and basis, population, exclusions, rule...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
EVD-006-02	Governance, scope, and change: verify dimensional consistency, structure mapping, comparator behavior, boundary behavior, exclusions, and conflict-policy execution	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
EVD-006-03	Human authority and approval: include versioned positive, negative, boundary, and conflict test cases with approved expected results	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-007-01	Human authority and approval: implement proposal, independent clinical review, approval, publication, periodic review, retirement, emergency suspension, and rollback...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-007-02	Human authority and approval: be available in clinical mode, and rollback shall restore a previously approved version without...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-007-03	Human authority and approval: retain the actor, role, rationale, source change, impact assessment, approval, effective time, and affected...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-008-01	Evidence and clinical rules: return an explicit no-source or no-applicable-rule status when it cannot produce a verified rule...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-008-02	Human authority and approval: block dependent automated reasoning, identify the missing evidence need, and route the case to...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
EVD-008-03	Human authority and approval: record the query, evaluated sources, exclusion reasons, no-rule status, human resolution, and any approved...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.5 Clinical context, imaging, registration, and contours (CLN) - 27 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
CLN-001-01	Human authority and approval: derive candidate workflow matches only from verified case facts and approved workflow definitions	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-001-02	Human factors and use safety: display candidate rationale and missing material information, shall require clinician confirmation, and shall not...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
CLN-001-03	Data integrity, provenance, and audit: record the verified inputs, workflow candidates, missing-data findings, clinician selection or rejection, and corrections	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
CLN-002-01	Anatomy, registration, and geometry: identify the approved image studies, frames of reference, registrations, structure sets, prior-treatment information, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-002-02	Verification, commissioning, and release: reject a planning task with a missing, conflicting, stale, or incorrectly referenced required input	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-002-03	Anatomy, registration, and geometry: preserve the validated planning-context manifest and bind it to every derived anatomy, intent, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-003-01	Prescription, plan, and dose: allow only commissioned rigid-registration, deformable-registration, and dose-accumulation methods for the active anatomy, image conditions...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
CLN-003-02	Prescription, plan, and dose: provide registration-specific visualization, quality measures, uncertainty documentation, and an authorized approval step before derived...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
CLN-003-03	Prescription, plan, and dose: retain the source and target images, transform, method and version, settings, quality results, uncertainty...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
CLN-004-01	Anatomy, registration, and geometry: verify the approved anatomy, imaging conditions, structure list, patient population, use case, model version...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-004-02	Anatomy, registration, and geometry: be created in a draft state and shall retain its model, input, configuration, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-004-03	Anatomy, registration, and geometry: prohibit a model or automated service from approving a generated contour or promoting it...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-005-01	Anatomy, registration, and geometry: calculate the commissioned uncertainty, input-quality, anomaly, and out-of-distribution indicators defined for the active model...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-005-02	Anatomy, registration, and geometry: abstain or route to manual contouring when an approved applicability threshold is not met...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
CLN-005-03	Verification, commissioning, and release: record all applicability indicators, thresholds, decisions, user responses, and performance against commissioned abstinence and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-006-01	Anatomy, registration, and geometry: evaluate structure identity, approved nomenclature mapping, laterality, required presence, completeness, topology, image coverage, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-006-02	Anatomy, registration, and geometry: block promotion to clinical-candidate status while any required contour validation is failed, unresolved, or...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-006-03	Evidence and clinical rules: provide and retain a validation report identifying each check, result, affected structure, evidence, disposition...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-007-01	Anatomy, registration, and geometry: reference its input study, frame of reference, model and version, configuration, naming map, confidence...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
CLN-007-02	Anatomy, registration, and geometry: create a new contour version and shall retain the changed region, editor, time, reason...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-007-03	Anatomy, registration, and geometry: identify the exact contour version, lineage, reviewer, approval state, and integrity value transferred downstream	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
CLN-008-01	Anatomy, registration, and geometry: provide synchronized slice, surface, difference, and configured clinically relevant error views for the selected...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-008-02	Anatomy, registration, and geometry: capture the location, spatial extent, operation type, and before-and-after geometry for each user edit	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-008-03	Anatomy, registration, and geometry: verify that representative users can find, interpret, edit, and confirm clinically important contour errors...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-009-01	Human authority and approval: identify the authorized approval role for targets, organs at risk, derived structures, and each...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-009-02	Human authority and approval: require radiation-oncologist approval for clinical targets and shall enforce the configured authorized role before...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
CLN-009-03	Anatomy, registration, and geometry: bind the approver, role, contour version, review time, approval decision, and any conditions, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.6 Treatment-plan generation and calculation (PLN) - 36 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
PLN-001-01	Prescription, plan, and dose: reference the signed prescription, approved anatomy, patient-specific instructions, applicable protocol, institutional planning standard, and...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-001-02	Prescription, plan, and dose: identify missing inputs and conflicts among authoritative sources and shall block candidate generation until...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-001-03	Data integrity, provenance, and audit: retain its source versions, normalized content, assumptions, clarifications, author or confirmer, integrity value, and...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-002-01	Prescription, plan, and dose: list each commissioned planning, optimization, robustness, and dose-calculation service with its validated machine, technique...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-002-02	Prescription, plan, and dose: verify the active request and configuration against the commissioned-use envelope before invoking a clinical...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-002-03	Human authority and approval: record the service identity, version, approved configuration, envelope decision, input identity, and output identity...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-003-01	Prescription, plan, and dose: separate language, speech, disease-site, and generative-model services from commissioned numerical dose-calculation components and credentials	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
PLN-003-02	Prescription, plan, and dose: accept clinical dose only from an approved commissioned dose service and shall reject model-supplied...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-003-03	Prescription, plan, and dose: demonstrate rejection of spoofed, mislabeled, malformed, or model-generated dose results at every supported interface...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-004-01	Prescription, plan, and dose: have an explicit type of hard constraint, planning goal, optimization objective, preference, or reporting-only...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-004-02	Human factors and use safety: display criterion type, priority, units, source, and status without presenting a preference or reporting...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-004-03	Prescription, plan, and dose: apply behavior consistent with the declared criterion type and shall prevent implicit type conversion	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-005-01	Governance, scope, and change: prohibit modification or relaxation of a hard constraint unless the active policy permits it...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-005-02	Prescription, plan, and dose: require an explicit action, structured rationale, and a pre-change and post-change difference review before...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-005-03	Human authority and approval: create a new planning-intent version and shall retain the actor, authority, rationale, old and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
PLN-006-01	Prescription, plan, and dose: contain the authorized requested candidate count and shall assign a unique identity to each...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-006-02	Prescription, plan, and dose: report generated, failed, infeasible, canceled, and pending candidates and shall state why the requested...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-006-03	Operations, resilience, and monitoring: retain the requested count, generation attempts, parameters, job states, results, and user decision to...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-007-01	Prescription, plan, and dose: identify the planning strategy and the parameter, objective, beam, modality, robustness, or other controlled...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-007-02	Human authority and approval: calculate and present configured diversity descriptors against the parent candidate set without claiming clinically...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-007-03	Data integrity, provenance, and audit: retain the diversity method, configuration, descriptors, related candidates, and any user-directed variation	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-008-01	Data integrity, provenance, and audit: use versioned feasibility rules and declared dominance criteria appropriate to the active modality, technique...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-008-02	Prescription, plan, and dose: label infeasible candidates and candidates dominated under the declared criteria and shall expose the...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
PLN-008-03	Human authority and approval: retain rule versions, input metrics, comparator results, excluded candidates, and any authorized human disposition	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-009-01	Prescription, plan, and dose: expose a signed runtime manifest containing machine model, energy or beam type, technique, dose...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-009-02	Human authority and approval: compare the runtime manifest with the approved release configuration and shall block calculation on...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-009-03	Prescription, plan, and dose: be bound to each calculation output and candidate-plan version	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-010-01	Prescription, plan, and dose: identify numeric units, physical or RBE-weighted basis, fractionation, normalization, biological transformation when approved, and...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-010-02	Prescription, plan, and dose: prevent comparison, summation, or conversion of incompatible dose quantities unless an approved explicit transformation...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-010-03	Prescription, plan, and dose: present the dose basis and assumptions adjacent to each value and shall retain every...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
PLN-011-01	Prescription, plan, and dose: define the required setup, range, motion, interplay, and other robustness scenarios, including parameter values...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
PLN-011-02	Human authority and approval: execute every required scenario and shall block candidate eligibility when a scenario is missing...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
PLN-011-03	Human factors and use safety: retain and present each scenario definition, result, aggregate, warning, and eligibility decision	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-012-01	Governance, scope, and change: have a unique immutable identity and shall reference its parent intent, predecessor candidate, parameters...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-012-02	Human factors and use safety: create a new candidate version without overwriting prior parameters, results, warnings, evaluation states, or...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
PLN-012-03	Data integrity, provenance, and audit: be able to compare any retained candidate versions and reconstruct the ordered refinement history...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

6.7 Plan review, selection, QA, and transfer (REV) - 27 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
REV-001-01	Human factors and use safety: use one versioned metric definition, unit convention, calculation method, and default display scale for...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
REV-001-02	Human authority and approval: present absolute values and candidate-to-reference differences and shall visibly identify any authorized scale or...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
REV-001-03	Data integrity, provenance, and audit: retain the candidate set, metric-definition version, scales, reference candidate, filters, and user changes	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
REV-002-01	Prescription, plan, and dose: provide target, organ-at-risk, constraint, spatial dose, hotspot, coldspot, robustness, uncertainty, complexity, deliverability, and provenance...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
REV-002-02	Human authority and approval: identify missing or stale required review content and shall block promotion until the configured...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-002-03	Human factors and use safety: link to the exact candidate, calculation, definition, scenario, and provenance record from which it...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
REV-003-01	Human factors and use safety: display each declared trade-off criterion, value, priority or weight, and candidate-specific contribution used in...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
REV-003-02	Human authority and approval: not automatically select a final candidate and shall not present an unexplained composite score...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-003-03	AI model and inference: include an inspectable rationale, uncertainty and limitations, and the audit record shall identify the...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-004-01	Interoperability and external TPS: identify the selected candidate, rejected alternatives, reviewer roles, comments, deviations, tradeoff rationale, and decision...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
REV-004-02	Verification, commissioning, and release: require completion of configured decision fields and role-specific attestations before the review can be...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-004-03	Data integrity, provenance, and audit: be integrity protected, version linked, time stamped, and retained with all subsequent amendments or...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
REV-005-01	Human authority and approval: assign clinical plan approval to an authorized radiation oncologist and technical review to an...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-005-02	Human authority and approval: require both current role-specific decisions and shall enforce configured ordering and separation-of-duties constraints before...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-005-03	Human authority and approval: bind the actor, role, candidate version, reviewed evidence, decision, conditions, time, and authentication event	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-006-01	Prescription, plan, and dose: define required independent dose or MU verification, patient-specific QA, transfer QA, and technique-specific checks...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
REV-006-02	Human authority and approval: create and track every required QA task and shall block release while a required...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
REV-006-03	Data integrity, provenance, and audit: retain each QA input, method, result, tolerance, reviewer, disposition, linked candidate version, and supersession...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
REV-007-01	Human authority and approval: verify that all required approvals and QA states are current for the exact source-object...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-007-02	Anatomy, registration, and geometry: compare patient and object identity, content, units, geometry, references, integrity values, and destination acknowledgment...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-007-03	Interoperability and external TPS: retain the source package, destination, transmitted objects, validation results, acknowledgments, failures, retries, and reconciliation...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-008-01	Human authority and approval: be immutable; any authorized change shall create a new version linked to the approved...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-008-02	Human authority and approval: identify and invalidate every downstream approval, QA result, transfer status, and derived object affected...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-008-03	Data integrity, provenance, and audit: present the required re-review set and shall retain each completed re-review and resulting restoration...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
REV-009-01	Human authority and approval: have a controlled independence record identifying its owner, purpose, inputs, algorithms, dependencies, data sources...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-009-02	AI model and inference: evaluate shared inputs, common software, models, configurations, infrastructure, personnel, and failure modes and shall...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
REV-009-03	Verification, commissioning, and release: expose material independence limitations, and the system shall retain the assessment version used for...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.8 Data, interoperability, provenance, and audit (DAT) - 30 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
DAT-001-01	Interoperability and external TPS: require a globally unique identity, version, lifecycle state, owner, creation source, and source linkage...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-001-02	Data integrity, provenance, and audit: reject identifier collision, missing ownership, invalid lifecycle transition, or creation of a derived object...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
DAT-001-03	Data integrity, provenance, and audit: expose the identity, version, state, owner, and source linkage for each clinical object	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
DAT-002-01	Anatomy, registration, and geometry: represent typed edges from every derived object to its input objects, transformations, software, models...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
DAT-002-02	Data integrity, provenance, and audit: prevent promotion of an object with a missing mandatory provenance edge, orphaned dependency, or...	R	3/3/3/3/5; 3.20	4/4/4/4/3; 3.90	4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
DAT-002-03	Data integrity, provenance, and audit: be able to query transitive upstream and downstream provenance and reproduce the dependency set...	R	3/3/3/3/5; 3.20	4/4/4/4/3; 3.90	4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
DAT-003-01	Prescription, plan, and dose: define canonical units, coordinate systems, dose quantities and bases, structure definitions, and rounding rules...	R	1/3/2/2/5; 2.15	4/4/4/4/3; 3.90	5/4/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
DAT-003-02	Data integrity, provenance, and audit: be explicit, versioned, dimensionally validated, and rejected when the source or destination representation is...	R	3/3/3/3/5; 3.20	4/4/4/4/3; 3.90	4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
DAT-003-03	Anatomy, registration, and geometry: cover storage, calculation, display, export, round-trip, precision, rounding, coordinate-transform, and incompatible-unit cases	R	2/3/2/2/5; 2.60	4/4/4/4/3; 3.90	5/4/4/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-004-01	Anatomy, registration, and geometry: validate patient and study identity, SOP Instance UIDs, object references, frame-of-reference UIDs, required attributes...	R	2/3/2/2/5; 2.60	4/4/4/4/3; 3.90	5/4/4/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-004-02	Data integrity, provenance, and audit: reject or quarantine a transfer containing a missing, duplicate, inconsistent, orphaned, or destination-altered required...	R	3/3/3/3/5; 3.20	4/4/4/4/3; 3.90	4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
DAT-004-03	Anatomy, registration, and geometry: compare source and destination identities, references, content, geometry, units, integrity values, and acknowledgment status	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-005-01	Human authority and approval: list the supported DICOM SOP classes, roles, transfer syntaxes, transactions, required attributes, and known...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-005-02	Human authority and approval: import, validate, create, and export only the DICOM objects and transactions approved for the...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-005-03	Prescription, plan, and dose: use representative and challenge objects for images, registrations, segmentations or structures, plans, ion plans...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
DAT-006-01	Interoperability and external TPS: identify the profile or guide version, actor or resource role, supported transactions, terminology, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-006-02	Human authority and approval: validate conformance to the approved profile or guide and shall prevent use of an...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-006-03	Data integrity, provenance, and audit: maintain and make available a conformance statement and known-limitations record for every implemented exchange	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
DAT-007-01	Verification, commissioning, and release: classify missing, ambiguous, inconsistent, orphaned, duplicate, and partial-transfer conditions with explicit invalid or quarantine...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-007-02	Human authority and approval: prevent invalid or quarantined data from becoming active, approved, or eligible as input to...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-007-03	Interoperability and external TPS: identify each detected condition, affected object, source, disposition, correction, and reconciliation result	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-008-01	Data integrity, provenance, and audit: use append-only records, trusted time, ordered event identity, actor identity, object versions, action type...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
DAT-008-02	Human authority and approval: capture commands, context, inputs, versions, evidence, outputs, warnings, edits, approvals, QA, exports, failures, overrides...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-008-03	Cybersecurity and privacy: preserve event order and integrity while enforcing access control, retention, legal hold, and privacy...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-009-01	Human authority and approval: create and retain an integrity hash, digital signature, or approved equivalent for each protected...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-009-02	Human authority and approval: be verified at approved creation, load, use, transfer, restoration, and reconciliation boundaries, with failure...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
DAT-009-03	Cybersecurity and privacy: create a security and clinical-safety event containing the artifact, expected and observed values, context...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
DAT-010-01	Data integrity, provenance, and audit: identify the authoritative system of record, retention period, legal-record status, export policy, archive, and...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
DAT-010-02	Governance, scope, and change: enforce the active information-governance matrix and applicable legal holds	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
DAT-010-03	Evidence and clinical rules: compare authoritative and dependent records and shall retain discrepancies, corrective actions, and closure evidence	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.9 AI and inference-model lifecycle (AIM) - 24 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
AIM-001-01	Human authority and approval: identify each approved model version, task, intended use, operating modes, clinical scope, artifact signature...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-001-02	Human authority and approval: invoke only the exact locked artifact approved for the active task, scope, and operating...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-001-03	Human authority and approval: retain the requested and resolved model identities, registry version, signature result, approval status, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
AIM-002-01	Anatomy, registration, and geometry: contain a machine-readable input contract for intended use, population, data or imaging conditions, required...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-002-02	AI model and inference: validate the active case and prepared inputs against every applicable input-contract condition before inference...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-002-03	Data integrity, provenance, and audit: expose and retain the input-contract results, preprocessing version, resource checks, rejected conditions, and routing...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
AIM-003-01	AI model and inference: define versioned abstention, rejection, and manual-review triggers for unsupported, anomalous, incomplete, and out-of-distribution inputs	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-003-02	Human authority and approval: produce no promotable clinical output and shall route the case to the configured manual...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-003-03	AI model and inference: measure and retain trigger performance, false acceptance, false rejection, routing outcome, and user disposition	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-004-01	AI model and inference: be read-only to clinical workflows and service identities	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-004-02	AI model and inference: not write to or update a production model or retrieval behavior through the clinical...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
AIM-004-03	Human authority and approval: occur in a segregated governed lifecycle with approved data use, evaluation, change control, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-005-01	Human authority and approval: require model version, intended use, owner, supplier, training and evaluation provenance, limitations, approved configuration...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-005-02	Anatomy, registration, and geometry: verify completeness, signatures, dependency identities, vulnerability status, and compatibility before the model becomes approved	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-005-03	AI model and inference: be queryable by authorized users and linked to the inference and resulting clinical object	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-006-01	AI model and inference: assign immutable identities and intended purposes to development, tuning, internal-test, external-test, and site-acceptance datasets	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-006-02	Data integrity, provenance, and audit: perform patient, study, source, temporal, and derived-data overlap checks and shall report all identified...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
AIM-006-03	Human authority and approval: receive documented approval before model release	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
AIM-007-01	Human authority and approval: define overall and subgroup performance, calibration, failure detection, uncertainty, edit burden, and known-limitation measures...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-007-02	AI model and inference: display the applicable model performance summary, limitations, uncertainty meaning, and commissioned action thresholds to...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-007-03	Human authority and approval: compare observed model and subgroup signals with approved thresholds and shall document resulting action	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-008-01	AI model and inference: classify changes to weights, behavior-altering prompts, retrieval configuration, clinical rules, preprocessing, dependencies, and runtime...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-008-02	Human authority and approval: receive documented impact assessment and the required regression, performance, safety, cybersecurity, and compatibility testing...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
AIM-008-03	Human authority and approval: create a new versioned release with an approved rollback plan and shall preserve the...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.10 Human factors, usability, and competency (HFE) - 18 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
HFE-001-01	Patient and case identity: display the active patient, course, study, plan, operating mode, approval state, and unsaved or...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
HFE-001-02	Human factors and use safety: require explicit transition handling and shall invalidate stale confirmations, previews, and unsaved actions associated...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-001-03	Human factors and use safety: verify context visibility and correct user response during interruptions, handoffs, multi-patient work, and version...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-002-01	Evidence and clinical rules: carry an explicit semantic source label	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
HFE-002-02	Human factors and use safety: use text, shape, iconography, or grouping in addition to color and shall remain consistent...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-002-03	Human factors and use safety: verify correct classification and interpretation of source types in normal, ambiguous, and time-pressured scenarios	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-003-01	Human factors and use safety: state the hazard or condition, affected patient or object, consequence when material, required action...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-003-02	Interoperability and external TPS: use accessible multimodal indicators, shall not rely on color alone, and shall require the...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
HFE-003-03	Human factors and use safety: record warning presentation and response, and usability review shall assess warning recognition, comprehension, and...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-004-01	Interoperability and external TPS: assign a risk tier, presentation method, acknowledgment requirement, escalation, suppression rule, and owner to...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
HFE-004-02	Interoperability and external TPS: collect alert frequency, repeated exposure, acknowledgment, override, cancellation, workaround, and response-time signals without weakening...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
HFE-004-03	Operations, resilience, and monitoring: periodically review alert burden and critical-warning comprehension and shall control corrective changes through risk...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-005-01	Operations, resilience, and monitoring: identify intended users, use environments, critical tasks, hazards, foreseeable misuse, interruptions, handoffs, recovery scenarios...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-005-02	Verification, commissioning, and release: document findings, risk links, design changes, unresolved issues, and verification of corrective actions	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
HFE-005-03	Human authority and approval: use representative users and conditions and shall demonstrate successful completion of critical tasks without...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
HFE-006-01	Human factors and use safety: define role-specific curricula, prerequisites, supervised practice, competency assessments, renewal intervals, and remediation for each...	R	3/3/3/3/5; 3.20	4/4/4/4/3; 3.90	4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-006-02	Human factors and use safety: enable a role-specific clinical function only when the user's training, competency, and authorization records...	R	3/3/3/3/5; 3.20	4/4/4/4/3; 3.90	4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
HFE-006-03	Evidence and clinical rules: retain training and competency evidence, authorization decisions, expirations, revocations, exceptions, and periodic review results	R	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.11 Privacy, cybersecurity, and secure development (SEC) - 24 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SEC-001-01	Human authority and approval: provide unique user identity, approved multi-factor authentication, role and context authorization, least privilege, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-001-02	Cybersecurity and privacy: evaluate current identity, role, context, session, and object authorization and shall reject absent, expired...	R	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-001-03	Human authority and approval: perform periodic access and privilege reviews and shall retain approvals, changes, dormant accounts, exceptions...	R	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SEC-002-01	Anatomy, registration, and geometry: use approved encryption in transit and at rest, network segmentation, managed secrets, endpoint controls...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-002-02	Human authority and approval: be continuously or periodically assessed against the approved baseline, with material deviation generating an...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-002-03	Human authority and approval: retain security-baseline versions, key and certificate status, configuration assessments, exceptions, approvals, and remediation evidence	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-003-01	Human authority and approval: allowlist approved connectors, model services, tenants, destinations, data classes, purposes, and required protection controls	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-003-02	Human authority and approval: block protected health information, images, prompts, transcripts, and outputs from an unapproved egress path...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-003-03	Operations, resilience, and monitoring: retain the source, destination, data classification, policy decision, actor, purpose, and incident escalation when...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SEC-004-01	Human authority and approval: enter the system through an untrusted-data boundary separate from system policy and tool authority	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SEC-004-02	Human authority and approval: sanitize, constrain, or reject embedded instructions that attempt to alter system authority, safety policy...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-004-03	Verification, commissioning, and release: cover prompt injection, indirect instruction, malformed metadata, encoded content, tool redirection, data exfiltration, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-005-01	Human authority and approval: have an approved manifest containing identity, version, provenance, integrity value, signature, and release status	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-005-02	Human authority and approval: occur before load or clinical use and shall block a missing, unsigned, altered, revoked...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-005-03	Verification, commissioning, and release: be retained for each artifact load	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-006-01	AI model and inference: require threat modeling, secure coding standards, code review, software bill of materials, vulnerability scanning...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-006-02	Human authority and approval: include objective evidence that required security activities are complete and that identified vulnerabilities are...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-006-03	Human authority and approval: undergo provenance, vulnerability, license, support, incident, and continuity assessment before approval	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SEC-007-01	Evidence and clinical rules: collect identity, access, export, integrity, connector, model, evidence, secret, and data-loss events with synchronized...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-007-02	Human authority and approval: identify unauthorized access, anomalous export, integrity failure, malicious input, artifact compromise, secrets exposure, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-007-03	Evidence and clinical rules: create a triage record containing severity, affected users and objects, clinical impact, containment, evidence...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-008-01	Evidence and clinical rules: define clinical and cybersecurity containment, evidence preservation, patient-impact assessment, communication, reporting, recovery, and corrective-action...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-008-02	Evidence and clinical rules: perform documented exercises covering representative compromise, outage, data-integrity, privacy, model, evidence, and interface incidents	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SEC-008-03	Evidence and clinical rules: retain chronology, decisions, affected cases, notifications, evidence, recovery validation, lessons learned, and corrective and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.12 Clinical operations, monitoring, and resilience (OPS) - 21 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OPS-001-01	Evidence and clinical rules: define each safety, model, evidence, workflow, technical, subgroup, reliability, and security signal with data...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-001-02	Human authority and approval: calculate signals against the approved thresholds and shall trigger the configured notification, restriction, suspension...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-001-03	Interoperability and external TPS: retain signal values, cohort or subgroup definition, threshold version, alert, action, acknowledgment, disposition, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-002-01	Data integrity, provenance, and audit: use a versioned state machine with pending, running, completed, failed, canceled, timed-out, partial, and...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OPS-002-02	Operations, resilience, and monitoring: be eligible for promotion; every other terminal or uncertain state shall remain ineligible and...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OPS-002-03	Data integrity, provenance, and audit: retain all job transitions, checkpoints, partial artifacts, cleanup actions, reconciliation decisions, and final eligibility	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OPS-003-01	Evidence and clinical rules: provide authorized suspension controls for a model, evidence rule, connector, workflow, disease site, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OPS-003-02	Human authority and approval: prevent new affected execution, identify in-progress and downstream affected work, and transition the service...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-003-03	Human authority and approval: restore a signed known-good configuration and shall require documented validation, reconciliation, approval, and communication...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-004-01	Operations, resilience, and monitoring: define allowed read functions, prohibited executions, authorized users, data availability, indicators, entry criteria, and...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OPS-004-02	Human authority and approval: display the condition and shall block new clinical execution, modification, approval, and export except...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-004-03	Operations, resilience, and monitoring: define access to completed work and audit records, external work practices, reconciliation, communication, and...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OPS-005-01	Human authority and approval: identify protected objects, versions, encryption, integrity checks, retention, recovery objectives, storage locations, and restoration...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-005-02	Human authority and approval: occur in a staged environment and shall verify object identity, dependencies, destination reconciliation, approval...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OPS-005-03	Human authority and approval: conduct and retain periodic recovery exercises, observed results, deviations, corrective actions, and approval for...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-006-01	Operations, resilience, and monitoring: support reports for incidents, near misses, unsafe workarounds, and QA discrepancies with severity, triage...	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OPS-006-02	Human authority and approval: link affected patients, cases, objects, releases, models, evidence, interfaces, and prior related events while...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-006-03	Human authority and approval: document patient impact, evidence preservation, root or contributing causes, trends, corrective and preventive action...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-007-01	Human authority and approval: define approved measures, thresholds, owners, escalation, and review intervals for availability, latency, recovery, retention...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-007-02	Human authority and approval: measure each service-level indicator using approved definitions and shall notify the responsible owner when...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OPS-007-03	Human authority and approval: retain performance trends, threshold breaches, actions, exceptions, proposed threshold changes, and governance approval	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

6.13 Verification, commissioning, release, and change control (VAL) - 27 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
VAL-001-01	Human authority and approval: assign stable identities and typed links among stakeholder needs, hazards, risk controls, high-level requirements...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-001-02	Verification, commissioning, and release: identify missing, orphaned, duplicate, circular, stale, or failed links and shall prevent baseline acceptance...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-001-03	Human authority and approval: retain an approved traceability report, anomaly dispositions, coverage results, and the configuration of all...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-002-01	Human authority and approval: define acceptance, commissioning, end-to-end, regression, shadow-mode, and governance criteria for the approved site configuration...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-002-02	Human authority and approval: execute the approved site test protocols using controlled configurations, expected results, tolerances, independent reviewers...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-002-03	Human authority and approval: remain disabled until all required site evidence is complete, anomalies are acceptably dispositioned, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-003-01	Prescription, plan, and dose: contain locked reference and challenge cases for language, voice, evidence, segmentation, registration, planning, dose...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
VAL-003-02	Data integrity, provenance, and audit: include immutable inputs, expected results or adjudication rules, configuration, applicability, provenance, and acceptance criteria	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
VAL-003-03	Governance, scope, and change: map cases to requirements, hazards, boundaries, failure modes, supported configurations, and known residual gaps	R	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
VAL-004-01	Human authority and approval: include normal, boundary, rare but foreseeable, wrong-context, partial-failure, time-out, rollback, post-approval change, and recovery...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-004-02	Human authority and approval: execute through the official user, model, clinical-service, and interface paths using the release-candidate configuration	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-004-03	Evidence and clinical rules: retain ordered evidence, expected and observed behavior, data lineage, screenshots or logs when applicable...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-005-01	Prescription, plan, and dose: define task-specific technical performance, clinically significant error, dose impact where relevant, edit burden, failure...	R	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
VAL-005-02	Human authority and approval: use representative and challenge data, pre-specified analyses, independent adjudication, confidence intervals or uncertainty characterization...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
VAL-005-03	Human authority and approval: document limitations, subgroup findings, user interaction effects, residual risk, monitoring needs, and approval for...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-006-01	Human authority and approval: have an approved protocol defining scope, endpoints, stop rules, sample rationale, adjudication, monitoring, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-006-02	Cybersecurity and privacy: obtain and retain all applicable research, ethics, privacy, quality, institutional, and regulatory authorizations before...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-006-03	Human authority and approval: require independent endpoint adjudication, anomaly and patient-impact review, residual-risk assessment, and documented approval to...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-007-01	Human authority and approval: have a versioned checklist of required safety, quality, subgroup, usability, reliability, cybersecurity, clinical, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-007-02	Human authority and approval: prevent progression or clinical activation when required evidence is missing, failed, expired, inconsistent, or...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-007-03	Human authority and approval: retain submitted evidence versions, automated checks, reviewer decisions, exceptions, residual risks, approval times, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
VAL-008-01	Evidence and clinical rules: classify expansions by disease site, model, modality, machine, technique, institution, adaptive workflow, interface, and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-008-02	Cybersecurity and privacy: prescribe the required commissioning, regression, human-factors, interoperability, cybersecurity, clinical, and monitoring activities for the...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-008-03	Human authority and approval: remain unavailable in clinical mode until all prescribed activities and approvals are complete and...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-009-01	Human authority and approval: identify severity, clinical impact, affected scope, risk-control status, release-threshold effect, owner, disposition, and closure...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-009-02	Human authority and approval: block entry to or continued operation in clinical mode when a critical anomaly, unacceptable...	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
VAL-009-03	Human authority and approval: require containment, affected-case assessment, corrective action, regression evidence, residual-risk approval, and documented authorization	R	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7 High-level interface requirement trade matrices

7.1 Governance, intended use, and scope (GOV) - 21 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-GOV-001-01	Governance, scope, and change: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-001-02	Governance, scope, and change: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-001-03	Governance, scope, and change: maintain a versioned trace from GOV-001 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-002-01	Governance, scope, and change: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-002-02	Governance, scope, and change: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-002-03	Governance, scope, and change: maintain a versioned trace from GOV-002 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-003-01	Governance, scope, and change: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-GOV-003-02	Governance, scope, and change: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-003-03	Governance, scope, and change: maintain a versioned trace from GOV-003 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-004-01	Governance, scope, and change: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-004-02	Governance, scope, and change: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-004-03	Governance, scope, and change: maintain a versioned trace from GOV-004 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-005-01	Governance, scope, and change: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-005-02	Governance, scope, and change: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-GOV-005-03	Governance, scope, and change: maintain a versioned trace from GOV-005 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-006-01	Governance, scope, and change: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-006-02	Governance, scope, and change: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-006-03	Governance, scope, and change: maintain a versioned trace from GOV-006 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-007-01	Governance, scope, and change: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-007-02	Governance, scope, and change: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-GOV-007-03	Governance, scope, and change: maintain a versioned trace from GOV-007 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

7.2 Safety boundaries and human authority (SAF) - 30 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SAF-001-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-001-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-SAF-001-03	Human authority and approval: maintain a versioned trace from SAF-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-002-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-002-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-SAF-002-03	Human authority and approval: maintain a versioned trace from SAF-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-003-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-003-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SAF-003-03	Human authority and approval: maintain a versioned trace from SAF-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-004-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-004-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-SAF-004-03	Human authority and approval: maintain a versioned trace from SAF-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-005-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-005-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-SAF-005-03	Human authority and approval: maintain a versioned trace from SAF-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-006-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SAF-006-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-SAF-006-03	Human authority and approval: maintain a versioned trace from SAF-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-007-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-007-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-SAF-007-03	Human authority and approval: maintain a versioned trace from SAF-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-008-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-008-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SAF-008-03	Human authority and approval: maintain a versioned trace from SAF-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-009-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-009-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-SAF-009-03	Human authority and approval: maintain a versioned trace from SAF-009 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-010-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SAF-010-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-SAF-010-03	Human authority and approval: maintain a versioned trace from SAF-010 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.3 Natural-language and voice interaction (NLI) - 24 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-NLI-001-01	Natural language and intent: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-001-02	Natural language and intent: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-001-03	Natural language and intent: maintain a versioned trace from NLI-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-002-01	Natural language and intent: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-002-02	Natural language and intent: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-002-03	Natural language and intent: maintain a versioned trace from NLI-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-003-01	Natural language and intent: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-003-02	Natural language and intent: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-NLI-003-03	Natural language and intent: maintain a versioned trace from NLI-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-004-01	Natural language and intent: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-004-02	Natural language and intent: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-004-03	Natural language and intent: maintain a versioned trace from NLI-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-005-01	Natural language and intent: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-005-02	Natural language and intent: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-005-03	Natural language and intent: maintain a versioned trace from NLI-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-006-01	Natural language and intent: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-NLI-006-02	Natural language and intent: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-006-03	Natural language and intent: maintain a versioned trace from NLI-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-007-01	Natural language and intent: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-007-02	Natural language and intent: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-007-03	Natural language and intent: maintain a versioned trace from NLI-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-008-01	Natural language and intent: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-NLI-008-02	Natural language and intent: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-NLI-008-03	Natural language and intent: maintain a versioned trace from NLI-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.4 Evidence and literature governance (EVD) - 24 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-EVD-001-01	Evidence and clinical rules: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-001-02	Evidence and clinical rules: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-001-03	Evidence and clinical rules: maintain a versioned trace from EVD-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-002-01	Evidence and clinical rules: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-002-02	Evidence and clinical rules: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-EVD-002-03	Evidence and clinical rules: maintain a versioned trace from EVD-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-003-01	Evidence and clinical rules: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-003-02	Evidence and clinical rules: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-003-03	Evidence and clinical rules: maintain a versioned trace from EVD-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-004-01	Evidence and clinical rules: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-004-02	Evidence and clinical rules: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-004-03	Evidence and clinical rules: maintain a versioned trace from EVD-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-005-01	Evidence and clinical rules: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-EVD-005-02	Evidence and clinical rules: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-005-03	Evidence and clinical rules: maintain a versioned trace from EVD-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-006-01	Evidence and clinical rules: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-006-02	Evidence and clinical rules: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-006-03	Evidence and clinical rules: maintain a versioned trace from EVD-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-007-01	Evidence and clinical rules: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-007-02	Evidence and clinical rules: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-EVD-007-03	Evidence and clinical rules: maintain a versioned trace from EVD-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-008-01	Evidence and clinical rules: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-008-02	Evidence and clinical rules: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-EVD-008-03	Evidence and clinical rules: maintain a versioned trace from EVD-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.5 Clinical context, imaging, registration, and contours (CLN) - 27 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-CLN-001-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-001-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-CLN-001-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-002-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-002-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-CLN-002-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-003-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-003-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-CLN-003-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-CLN-004-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-004-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-CLN-004-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-005-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-005-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-CLN-005-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-006-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-CLN-006-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-CLN-006-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-007-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-007-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-CLN-007-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-008-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-008-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-CLN-008-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-009-01	Anatomy, registration, and geometry: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-CLN-009-02	Anatomy, registration, and geometry: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-CLN-009-03	Anatomy, registration, and geometry: maintain a versioned trace from CLN-009 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.6 Treatment-plan generation and calculation (PLN) - 36 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-PLN-001-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-001-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-PLN-001-03	Prescription, plan, and dose: maintain a versioned trace from PLN-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-002-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-002-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-002-03	Prescription, plan, and dose: maintain a versioned trace from PLN-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-003-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-003-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-003-03	Prescription, plan, and dose: maintain a versioned trace from PLN-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-004-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-PLN-004-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-004-03	Prescription, plan, and dose: maintain a versioned trace from PLN-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-005-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-005-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-005-03	Prescription, plan, and dose: maintain a versioned trace from PLN-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-006-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-006-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-PLN-006-03	Prescription, plan, and dose: maintain a versioned trace from PLN-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-007-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-007-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-007-03	Prescription, plan, and dose: maintain a versioned trace from PLN-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-008-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-008-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-008-03	Prescription, plan, and dose: maintain a versioned trace from PLN-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-009-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-PLN-009-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-009-03	Prescription, plan, and dose: maintain a versioned trace from PLN-009 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-010-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-010-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-010-03	Prescription, plan, and dose: maintain a versioned trace from PLN-010 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-011-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-011-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-PLN-011-03	Prescription, plan, and dose: maintain a versioned trace from PLN-011 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-012-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-PLN-012-02	Prescription, plan, and dose: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-PLN-012-03	Prescription, plan, and dose: maintain a versioned trace from PLN-012 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.7 Plan review, selection, QA, and transfer (REV) - 27 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-REV-001-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-001-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-REV-001-03	Human authority and approval: maintain a versioned trace from REV-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-002-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-002-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-REV-002-03	Human authority and approval: maintain a versioned trace from REV-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-003-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-003-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-REV-003-03	Human authority and approval: maintain a versioned trace from REV-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-004-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-REV-004-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-REV-004-03	Human authority and approval: maintain a versioned trace from REV-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-005-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-005-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-REV-005-03	Human authority and approval: maintain a versioned trace from REV-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-006-01	Prescription, plan, and dose: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-006-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-REV-006-03	Human authority and approval: maintain a versioned trace from REV-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-007-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-007-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-REV-007-03	Human authority and approval: maintain a versioned trace from REV-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-008-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-008-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-REV-008-03	Human authority and approval: maintain a versioned trace from REV-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-REV-009-01	Human authority and approval: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-REV-009-02	Human authority and approval: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-REV-009-03	Human authority and approval: maintain a versioned trace from REV-009 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.8 Data, interoperability, provenance, and audit (DAT) - 30 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-DAT-001-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-001-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-001-03	Interoperability and external TPS: maintain a versioned trace from DAT-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-002-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-DAT-002-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-002-03	Interoperability and external TPS: maintain a versioned trace from DAT-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-003-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-003-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-003-03	Interoperability and external TPS: maintain a versioned trace from DAT-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-004-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-004-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-DAT-004-03	Interoperability and external TPS: maintain a versioned trace from DAT-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-005-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-005-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-005-03	Interoperability and external TPS: maintain a versioned trace from DAT-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-006-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-006-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-006-03	Interoperability and external TPS: maintain a versioned trace from DAT-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-DAT-007-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-007-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-007-03	Interoperability and external TPS: maintain a versioned trace from DAT-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-008-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-008-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-008-03	Interoperability and external TPS: maintain a versioned trace from DAT-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-009-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-DAT-009-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-009-03	Interoperability and external TPS: maintain a versioned trace from DAT-009 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-010-01	Interoperability and external TPS: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-DAT-010-02	Interoperability and external TPS: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IF-DAT-010-03	Interoperability and external TPS: maintain a versioned trace from DAT-010 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.9 AI and inference-model lifecycle (AIM) - 24 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-AIM-001-01	AI model and inference: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-AIM-001-02	AI model and inference: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-001-03	AI model and inference: maintain a versioned trace from AIM-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-002-01	AI model and inference: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-002-02	AI model and inference: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-002-03	AI model and inference: maintain a versioned trace from AIM-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-003-01	AI model and inference: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-003-02	AI model and inference: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-AIM-003-03	AI model and inference: maintain a versioned trace from AIM-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-004-01	AI model and inference: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-004-02	AI model and inference: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-004-03	AI model and inference: maintain a versioned trace from AIM-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-005-01	AI model and inference: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-005-02	AI model and inference: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-005-03	AI model and inference: maintain a versioned trace from AIM-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-006-01	AI model and inference: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-AIM-006-02	AI model and inference: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-006-03	AI model and inference: maintain a versioned trace from AIM-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-007-01	AI model and inference: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-007-02	AI model and inference: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-007-03	AI model and inference: maintain a versioned trace from AIM-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-008-01	AI model and inference: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-AIM-008-02	AI model and inference: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-AIM-008-03	AI model and inference: maintain a versioned trace from AIM-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.10 Human factors, usability, and competency (HFE) - 18 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-HFE-001-01	Human factors and use safety: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-001-02	Human factors and use safety: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-001-03	Human factors and use safety: maintain a versioned trace from HFE-001 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-002-01	Human factors and use safety: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-002-02	Human factors and use safety: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-HFE-002-03	Human factors and use safety: maintain a versioned trace from HFE-002 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-003-01	Human factors and use safety: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-003-02	Human factors and use safety: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-003-03	Human factors and use safety: maintain a versioned trace from HFE-003 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-004-01	Human factors and use safety: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-004-02	Human factors and use safety: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-004-03	Human factors and use safety: maintain a versioned trace from HFE-004 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-005-01	Human factors and use safety: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-HFE-005-02	Human factors and use safety: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-005-03	Human factors and use safety: maintain a versioned trace from HFE-005 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-006-01	Human factors and use safety: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-006-02	Human factors and use safety: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-HFE-006-03	Human factors and use safety: maintain a versioned trace from HFE-006 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

7.11 Privacy, cybersecurity, and secure development (SEC) - 24 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SEC-001-01	Cybersecurity and privacy: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SEC-001-02	Cybersecurity and privacy: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-001-03	Cybersecurity and privacy: maintain a versioned trace from SEC-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-002-01	Cybersecurity and privacy: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-002-02	Cybersecurity and privacy: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-002-03	Cybersecurity and privacy: maintain a versioned trace from SEC-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-003-01	Cybersecurity and privacy: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-003-02	Cybersecurity and privacy: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SEC-003-03	Cybersecurity and privacy: maintain a versioned trace from SEC-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-004-01	Cybersecurity and privacy: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-004-02	Cybersecurity and privacy: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-004-03	Cybersecurity and privacy: maintain a versioned trace from SEC-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-005-01	Cybersecurity and privacy: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-005-02	Cybersecurity and privacy: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-005-03	Cybersecurity and privacy: maintain a versioned trace from SEC-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-006-01	Cybersecurity and privacy: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SEC-006-02	Cybersecurity and privacy: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-006-03	Cybersecurity and privacy: maintain a versioned trace from SEC-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-007-01	Cybersecurity and privacy: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-007-02	Cybersecurity and privacy: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-007-03	Cybersecurity and privacy: maintain a versioned trace from SEC-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-008-01	Cybersecurity and privacy: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-SEC-008-02	Cybersecurity and privacy: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-SEC-008-03	Cybersecurity and privacy: maintain a versioned trace from SEC-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

7.12 Clinical operations, monitoring, and resilience (OPS) - 21 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-OPS-001-01	Operations, re-silience, and monitoring: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-001-02	Operations, re-silience, and monitoring: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-001-03	Operations, re-silience, and monitoring: maintain a versioned trace from OPS-001 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-002-01	Operations, re-silience, and monitoring: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-002-02	Operations, re-silience, and monitoring: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-OPS-002-03	Operations, resilience, and monitoring: maintain a versioned trace from OPS-002 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-003-01	Operations, resilience, and monitoring: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-003-02	Operations, resilience, and monitoring: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-003-03	Operations, resilience, and monitoring: maintain a versioned trace from OPS-003 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-004-01	Operations, resilience, and monitoring: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-004-02	Operations, resilience, and monitoring: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-004-03	Operations, resilience, and monitoring: maintain a versioned trace from OPS-004 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-OPS-005-01	Operations, resilience, and monitoring: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-005-02	Operations, resilience, and monitoring: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-005-03	Operations, resilience, and monitoring: maintain a versioned trace from OPS-005 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-006-01	Operations, resilience, and monitoring: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-006-02	Operations, resilience, and monitoring: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-006-03	Operations, resilience, and monitoring: maintain a versioned trace from OPS-006 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-007-01	Operations, resilience, and monitoring: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-OPS-007-02	Operations, resilience, and monitoring: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IF-OPS-007-03	Operations, resilience, and monitoring: maintain a versioned trace from OPS-007 to interface-control documents, implementations, test cases, results, anomalies...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

7.13 Verification, commissioning, release, and change control (VAL) - 27 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-VAL-001-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-001-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-001-03	Verification, commissioning, and release: maintain a versioned trace from VAL-001 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-002-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-VAL-002-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-002-03	Verification, commissioning, and release: maintain a versioned trace from VAL-002 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-003-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-003-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-003-03	Verification, commissioning, and release: maintain a versioned trace from VAL-003 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-004-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-004-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-VAL-004-03	Verification, commissioning, and release: maintain a versioned trace from VAL-004 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-005-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-005-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-005-03	Verification, commissioning, and release: maintain a versioned trace from VAL-005 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-006-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-006-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-006-03	Verification, commissioning, and release: maintain a versioned trace from VAL-006 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-VAL-007-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-007-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-007-03	Verification, commissioning, and release: maintain a versioned trace from VAL-007 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-008-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-008-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-008-03	Verification, commissioning, and release: maintain a versioned trace from VAL-008 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-009-01	Verification, commissioning, and release: define the producer, consumer, authority, direction, trigger, data or object schema, required and conditional...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IF-VAL-009-02	Verification, commissioning, and release: validate identity, authorization, context, schema, version, references, units, integrity, and lifecycle state before acceptance...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IF-VAL-009-03	Verification, commissioning, and release: maintain a versioned trace from VAL-009 to interface-control documents, implementations, test cases, results, anomalies...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8 Sub-interface requirement trade matrices

8.1 Governance, intended use, and scope (GOV) - 63 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-001-01-01	Governance, scope, and change: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-001-01-02	Governance, scope, and change: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-001-01-03	Governance, scope, and change: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-001-02-01	Governance, scope, and change: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-001-02-02	Governance, scope, and change: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-001-02-03	Governance, scope, and change: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-001-03-01	Governance, scope, and change: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-001-03-02	Governance, scope, and change: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-001-03-03	Governance, scope, and change: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-002-01-01	Governance, scope, and change: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-002-01-02	Governance, scope, and change: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-002-01-03	Governance, scope, and change: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-002-02-01	Governance, scope, and change: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-002-02-02	Governance, scope, and change: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-002-02-03	Governance, scope, and change: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-002-03-01	Governance, scope, and change: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-002-03-02	Governance, scope, and change: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-002-03-03	Governance, scope, and change: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-003-01-01	Governance, scope, and change: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-003-01-02	Governance, scope, and change: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-003-01-03	Governance, scope, and change: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-003-02-01	Governance, scope, and change: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-003-02-02	Governance, scope, and change: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-003-02-03	Governance, scope, and change: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-003-03-01	Governance, scope, and change: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-003-03-02	Governance, scope, and change: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-003-03-03	Governance, scope, and change: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-004-01-01	Governance, scope, and change: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-004-01-02	Governance, scope, and change: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-004-01-03	Governance, scope, and change: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-004-02-01	Governance, scope, and change: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-004-02-02	Governance, scope, and change: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-004-02-03	Governance, scope, and change: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-004-03-01	Governance, scope, and change: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-004-03-02	Governance, scope, and change: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-004-03-03	Governance, scope, and change: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-005-01-01	Governance, scope, and change: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-005-01-02	Governance, scope, and change: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-005-01-03	Governance, scope, and change: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-005-02-01	Governance, scope, and change: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-005-02-02	Governance, scope, and change: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-005-02-03	Governance, scope, and change: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-005-03-01	Governance, scope, and change: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-005-03-02	Governance, scope, and change: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-005-03-03	Governance, scope, and change: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-006-01-01	Governance, scope, and change: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-006-01-02	Governance, scope, and change: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-006-01-03	Governance, scope, and change: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-006-02-01	Governance, scope, and change: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-006-02-02	Governance, scope, and change: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-006-02-03	Governance, scope, and change: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-006-03-01	Governance, scope, and change: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-006-03-02	Governance, scope, and change: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-006-03-03	Governance, scope, and change: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-007-01-01	Governance, scope, and change: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-007-01-02	Governance, scope, and change: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-007-01-03	Governance, scope, and change: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-007-02-01	Governance, scope, and change: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-007-02-02	Governance, scope, and change: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-GOV-007-02-03	Governance, scope, and change: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-007-03-01	Governance, scope, and change: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-007-03-02	Governance, scope, and change: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-GOV-007-03-03	Governance, scope, and change: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

8.2 Safety boundaries and human authority (SAF) - 90 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-001-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-001-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-001-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-001-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-001-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-001-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-001-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-001-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-001-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-002-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-002-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-002-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-002-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-002-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-002-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-002-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-002-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-002-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-003-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-003-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-003-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-003-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-003-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-003-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-003-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-003-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-003-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-004-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-004-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-004-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-004-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-004-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-004-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-004-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-004-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-004-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-005-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-005-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-005-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-005-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-005-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-005-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-005-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-005-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-005-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-006-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-006-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-006-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-006-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-006-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-006-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-006-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-006-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-006-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-007-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-007-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-007-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-007-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-007-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-007-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-007-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-007-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-007-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-008-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-008-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-008-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-008-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-008-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-008-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-008-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-008-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-008-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-009-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-009-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-009-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-009-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-009-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-009-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-009-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-009-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-009-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-010-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-010-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-010-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-010-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-010-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-010-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-SAF-010-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SAF-010-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SAF-010-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.3 Natural-language and voice interaction (NLI) - 72 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-001-01-01	Natural language and intent: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-001-01-02	Natural language and intent: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-001-01-03	Natural language and intent: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-001-02-01	Natural language and intent: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-001-02-02	Natural language and intent: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-001-02-03	Natural language and intent: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-001-03-01	Natural language and intent: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-001-03-02	Natural language and intent: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-001-03-03	Natural language and intent: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-002-01-01	Natural language and intent: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-002-01-02	Natural language and intent: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-002-01-03	Natural language and intent: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-002-02-01	Natural language and intent: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-002-02-02	Natural language and intent: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-002-02-03	Natural language and intent: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-002-03-01	Natural language and intent: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-002-03-02	Natural language and intent: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-002-03-03	Natural language and intent: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-003-01-01	Natural language and intent: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-003-01-02	Natural language and intent: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-003-01-03	Natural language and intent: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-003-02-01	Natural language and intent: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-003-02-02	Natural language and intent: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-003-02-03	Natural language and intent: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-003-03-01	Natural language and intent: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-003-03-02	Natural language and intent: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-003-03-03	Natural language and intent: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-004-01-01	Natural language and intent: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-004-01-02	Natural language and intent: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-004-01-03	Natural language and intent: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-004-02-01	Natural language and intent: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-004-02-02	Natural language and intent: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-004-02-03	Natural language and intent: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-004-03-01	Natural language and intent: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-004-03-02	Natural language and intent: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-004-03-03	Natural language and intent: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-005-01-01	Natural language and intent: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-005-01-02	Natural language and intent: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-005-01-03	Natural language and intent: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-005-02-01	Natural language and intent: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-005-02-02	Natural language and intent: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-005-02-03	Natural language and intent: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-005-03-01	Natural language and intent: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-005-03-02	Natural language and intent: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-005-03-03	Natural language and intent: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-006-01-01	Natural language and intent: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-006-01-02	Natural language and intent: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-006-01-03	Natural language and intent: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-006-02-01	Natural language and intent: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-006-02-02	Natural language and intent: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-006-02-03	Natural language and intent: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-006-03-01	Natural language and intent: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-006-03-02	Natural language and intent: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-006-03-03	Natural language and intent: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-007-01-01	Natural language and intent: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-007-01-02	Natural language and intent: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-007-01-03	Natural language and intent: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-007-02-01	Natural language and intent: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-007-02-02	Natural language and intent: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-007-02-03	Natural language and intent: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-007-03-01	Natural language and intent: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-007-03-02	Natural language and intent: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-007-03-03	Natural language and intent: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-008-01-01	Natural language and intent: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-008-01-02	Natural language and intent: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-008-01-03	Natural language and intent: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-008-02-01	Natural language and intent: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-008-02-02	Natural language and intent: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-NLI-008-02-03	Natural language and intent: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-008-03-01	Natural language and intent: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-008-03-02	Natural language and intent: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-NLI-008-03-03	Natural language and intent: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.4 Evidence and literature governance (EVD) - 72 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-001-01-01	Evidence and clinical rules: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-001-01-02	Evidence and clinical rules: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-001-01-03	Evidence and clinical rules: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-001-02-01	Evidence and clinical rules: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-001-02-02	Evidence and clinical rules: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-001-02-03	Evidence and clinical rules: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-001-03-01	Evidence and clinical rules: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-001-03-02	Evidence and clinical rules: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-001-03-03	Evidence and clinical rules: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-002-01-01	Evidence and clinical rules: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-002-01-02	Evidence and clinical rules: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-002-01-03	Evidence and clinical rules: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-002-02-01	Evidence and clinical rules: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-002-02-02	Evidence and clinical rules: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-002-02-03	Evidence and clinical rules: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-002-03-01	Evidence and clinical rules: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-002-03-02	Evidence and clinical rules: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-002-03-03	Evidence and clinical rules: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-003-01-01	Evidence and clinical rules: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-003-01-02	Evidence and clinical rules: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-003-01-03	Evidence and clinical rules: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-003-02-01	Evidence and clinical rules: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-003-02-02	Evidence and clinical rules: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-003-02-03	Evidence and clinical rules: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-003-03-01	Evidence and clinical rules: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-003-03-02	Evidence and clinical rules: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-003-03-03	Evidence and clinical rules: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-004-01-01	Evidence and clinical rules: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-004-01-02	Evidence and clinical rules: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-004-01-03	Evidence and clinical rules: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-004-02-01	Evidence and clinical rules: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-004-02-02	Evidence and clinical rules: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-004-02-03	Evidence and clinical rules: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-004-03-01	Evidence and clinical rules: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-004-03-02	Evidence and clinical rules: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-004-03-03	Evidence and clinical rules: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-005-01-01	Evidence and clinical rules: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-005-01-02	Evidence and clinical rules: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-005-01-03	Evidence and clinical rules: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-005-02-01	Evidence and clinical rules: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-005-02-02	Evidence and clinical rules: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-005-02-03	Evidence and clinical rules: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-005-03-01	Evidence and clinical rules: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-005-03-02	Evidence and clinical rules: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-005-03-03	Evidence and clinical rules: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-006-01-01	Evidence and clinical rules: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-006-01-02	Evidence and clinical rules: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-006-01-03	Evidence and clinical rules: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-006-02-01	Evidence and clinical rules: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-006-02-02	Evidence and clinical rules: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-006-02-03	Evidence and clinical rules: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-006-03-01	Evidence and clinical rules: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-006-03-02	Evidence and clinical rules: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-006-03-03	Evidence and clinical rules: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-007-01-01	Evidence and clinical rules: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-007-01-02	Evidence and clinical rules: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-007-01-03	Evidence and clinical rules: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-007-02-01	Evidence and clinical rules: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-007-02-02	Evidence and clinical rules: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-007-02-03	Evidence and clinical rules: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-007-03-01	Evidence and clinical rules: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-007-03-02	Evidence and clinical rules: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-007-03-03	Evidence and clinical rules: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-008-01-01	Evidence and clinical rules: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-008-01-02	Evidence and clinical rules: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-008-01-03	Evidence and clinical rules: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-008-02-01	Evidence and clinical rules: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-EVD-008-02-02	Evidence and clinical rules: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-008-02-03	Evidence and clinical rules: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-008-03-01	Evidence and clinical rules: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-008-03-02	Evidence and clinical rules: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-EVD-008-03-03	Evidence and clinical rules: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.5 Clinical context, imaging, registration, and contours (CLN) - 81 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-001-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-001-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-001-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-001-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-001-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-001-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-001-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-001-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-001-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-002-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-002-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-002-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-002-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-002-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-002-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-002-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-002-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-002-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-003-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-003-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-003-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-003-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-003-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-003-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-003-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-003-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-003-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-004-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-004-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-004-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-004-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-004-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-004-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-004-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-004-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-004-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-005-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-005-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-005-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-005-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-005-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-005-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-005-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-005-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-005-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-006-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-006-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-006-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-006-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-006-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-006-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-006-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-006-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-006-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-007-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-007-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-007-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-007-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-007-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-007-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-007-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-007-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-007-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-008-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-008-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-008-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-008-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-008-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-008-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-008-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-008-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-008-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-009-01-01	Anatomy, registration, and geometry: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-009-01-02	Anatomy, registration, and geometry: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-009-01-03	Anatomy, registration, and geometry: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-009-02-01	Anatomy, registration, and geometry: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-009-02-02	Anatomy, registration, and geometry: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-CLN-009-02-03	Anatomy, registration, and geometry: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-CLN-009-03-01	Anatomy, registration, and geometry: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-009-03-02	Anatomy, registration, and geometry: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-CLN-009-03-03	Anatomy, registration, and geometry: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.6 Treatment-plan generation and calculation (PLN) - 108 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-001-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-001-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-001-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-001-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-001-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-001-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-001-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-001-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-001-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-002-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-002-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-002-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-002-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-002-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-002-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-002-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-002-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-002-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-003-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-003-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-003-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-003-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-003-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-003-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-003-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-003-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-003-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-004-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-004-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-004-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-004-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-004-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-004-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-004-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-004-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-004-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-005-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-005-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-005-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-005-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-005-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-005-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-005-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-005-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-005-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-006-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-006-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-006-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-006-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-006-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-006-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-006-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-006-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-006-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-007-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-007-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-007-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-007-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-007-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-007-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-007-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-007-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-007-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-008-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-008-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-008-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-008-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-008-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-008-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-008-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-008-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-008-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-009-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-009-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-009-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-009-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-009-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-009-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-009-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-009-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-009-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-010-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-010-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-010-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-010-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-010-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-010-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-010-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-010-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-010-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-011-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-011-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-011-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-011-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-011-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-011-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-011-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-011-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-011-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-012-01-01	Prescription, plan, and dose: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-012-01-02	Prescription, plan, and dose: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-012-01-03	Prescription, plan, and dose: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-012-02-01	Prescription, plan, and dose: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-012-02-02	Prescription, plan, and dose: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-PLN-012-02-03	Prescription, plan, and dose: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-PLN-012-03-01	Prescription, plan, and dose: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-012-03-02	Prescription, plan, and dose: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-PLN-012-03-03	Prescription, plan, and dose: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.7 Plan review, selection, QA, and transfer (REV) - 81 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-001-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-001-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-001-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-001-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-001-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-001-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-001-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-001-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-001-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-002-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-002-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-002-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-002-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-002-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-002-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-002-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-002-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-002-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-003-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-003-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-003-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-003-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-003-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-003-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-003-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-003-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-003-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-004-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-004-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-004-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-004-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-004-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-004-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-004-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-004-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-004-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-005-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-005-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-005-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-005-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-005-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-005-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-005-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-005-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-005-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-006-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-006-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-006-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-006-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-006-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-006-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-006-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-006-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-006-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-007-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-007-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-007-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-007-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-007-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-007-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-007-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-007-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-007-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-008-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-008-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-008-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-008-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-008-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-008-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-008-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-008-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-008-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-009-01-01	Human authority and approval: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-009-01-02	Human authority and approval: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-REV-009-01-03	Human authority and approval: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-009-02-01	Human authority and approval: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-009-02-02	Human authority and approval: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-009-02-03	Human authority and approval: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-REV-009-03-01	Human authority and approval: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-009-03-02	Human authority and approval: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-REV-009-03-03	Human authority and approval: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.8 Data, interoperability, provenance, and audit (DAT) - 90 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-001-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-001-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-001-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-001-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-001-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-001-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-001-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-001-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-001-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-002-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-002-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-002-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-002-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-002-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-002-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-002-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-002-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-002-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-003-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-003-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-003-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-003-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-003-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-003-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-003-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-003-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-003-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-004-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-004-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-004-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-004-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-004-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-004-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-004-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-004-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-004-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-005-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-005-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-005-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-005-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-005-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-005-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-005-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-005-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-005-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-006-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-006-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-006-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-006-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-006-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-006-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-006-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-006-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-006-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-007-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-007-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-007-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-007-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-007-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-007-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-007-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-007-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-007-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-008-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-008-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-008-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-008-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-008-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-008-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-008-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-008-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-008-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-009-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-009-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-009-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-009-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-009-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-009-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-009-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-009-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-009-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-010-01-01	Interoperability and external TPS: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-010-01-02	Interoperability and external TPS: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-010-01-03	Data integrity, provenance, and audit: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-DAT-010-02-01	Interoperability and external TPS: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-010-02-02	Data integrity, provenance, and audit: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-010-02-03	Data integrity, provenance, and audit: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
SIR-DAT-010-03-01	Interoperability and external TPS: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-010-03-02	Interoperability and external TPS: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-DAT-010-03-03	Data integrity, provenance, and audit: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.9 AI and inference-model lifecycle (AIM) - 72 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-001-01-01	AI model and inference: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-001-01-02	AI model and inference: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-001-01-03	AI model and inference: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-001-02-01	AI model and inference: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-001-02-02	AI model and inference: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-001-02-03	AI model and inference: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-001-03-01	AI model and inference: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-001-03-02	AI model and inference: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-001-03-03	AI model and inference: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-002-01-01	AI model and inference: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-002-01-02	AI model and inference: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-002-01-03	AI model and inference: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-002-02-01	AI model and inference: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-002-02-02	AI model and inference: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-002-02-03	AI model and inference: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-002-03-01	AI model and inference: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-002-03-02	AI model and inference: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-002-03-03	AI model and inference: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-003-01-01	AI model and inference: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-003-01-02	AI model and inference: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-003-01-03	AI model and inference: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-003-02-01	AI model and inference: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-003-02-02	AI model and inference: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-003-02-03	AI model and inference: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-003-03-01	AI model and inference: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-003-03-02	AI model and inference: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-003-03-03	AI model and inference: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-004-01-01	AI model and inference: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-004-01-02	AI model and inference: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-004-01-03	AI model and inference: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-004-02-01	AI model and inference: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-004-02-02	AI model and inference: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-004-02-03	AI model and inference: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-004-03-01	AI model and inference: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-004-03-02	AI model and inference: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-004-03-03	AI model and inference: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-005-01-01	AI model and inference: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-005-01-02	AI model and inference: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-005-01-03	AI model and inference: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-005-02-01	AI model and inference: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-005-02-02	AI model and inference: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-005-02-03	AI model and inference: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-005-03-01	AI model and inference: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-005-03-02	AI model and inference: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-005-03-03	AI model and inference: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-006-01-01	AI model and inference: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-006-01-02	AI model and inference: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-006-01-03	AI model and inference: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-006-02-01	AI model and inference: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-006-02-02	AI model and inference: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-006-02-03	AI model and inference: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-006-03-01	AI model and inference: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-006-03-02	AI model and inference: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-006-03-03	AI model and inference: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-007-01-01	AI model and inference: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-007-01-02	AI model and inference: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-007-01-03	AI model and inference: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-007-02-01	AI model and inference: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-007-02-02	AI model and inference: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-007-02-03	AI model and inference: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-007-03-01	AI model and inference: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-007-03-02	AI model and inference: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-007-03-03	AI model and inference: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-008-01-01	AI model and inference: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-008-01-02	AI model and inference: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-AIM-008-01-03	AI model and inference: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-008-02-01	AI model and inference: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-008-02-02	AI model and inference: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-008-02-03	AI model and inference: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-008-03-01	AI model and inference: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-008-03-02	AI model and inference: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-AIM-008-03-03	AI model and inference: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.10 Human factors, usability, and competency (HFE) - 54 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-HFE-001-01-01	Human factors and use safety: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-001-01-02	Human factors and use safety: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-001-01-03	Human factors and use safety: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-001-02-01	Human factors and use safety: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-001-02-02	Human factors and use safety: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-001-02-03	Human factors and use safety: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-001-03-01	Human factors and use safety: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-HFE-001-03-02	Human factors and use safety: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-001-03-03	Human factors and use safety: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-002-01-01	Human factors and use safety: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-002-01-02	Human factors and use safety: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-002-01-03	Human factors and use safety: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-002-02-01	Human factors and use safety: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-002-02-02	Human factors and use safety: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-HFE-002-02-03	Human factors and use safety: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-002-03-01	Human factors and use safety: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-002-03-02	Human factors and use safety: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-002-03-03	Human factors and use safety: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-003-01-01	Human factors and use safety: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-003-01-02	Human factors and use safety: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-003-01-03	Human factors and use safety: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-HFE-003-02-01	Human factors and use safety: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-003-02-02	Human factors and use safety: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-003-02-03	Human factors and use safety: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-003-03-01	Human factors and use safety: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-003-03-02	Human factors and use safety: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-003-03-03	Human factors and use safety: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-004-01-01	Human factors and use safety: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-HFE-004-01-02	Human factors and use safety: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-004-01-03	Human factors and use safety: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-004-02-01	Human factors and use safety: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-004-02-02	Human factors and use safety: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-004-02-03	Human factors and use safety: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-004-03-01	Human factors and use safety: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-004-03-02	Human factors and use safety: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-HFE-004-03-03	Human factors and use safety: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-005-01-01	Human factors and use safety: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-005-01-02	Human factors and use safety: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-005-01-03	Human factors and use safety: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-005-02-01	Human factors and use safety: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-005-02-02	Human factors and use safety: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-005-02-03	Human factors and use safety: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-005-03-01	Human factors and use safety: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-HFE-005-03-02	Human factors and use safety: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-005-03-03	Human factors and use safety: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-006-01-01	Human factors and use safety: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-006-01-02	Human factors and use safety: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-006-01-03	Human factors and use safety: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-006-02-01	Human factors and use safety: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-006-02-02	Human factors and use safety: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-HFE-006-02-03	Human factors and use safety: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-006-03-01	Human factors and use safety: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-006-03-02	Human factors and use safety: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-HFE-006-03-03	Human factors and use safety: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

8.11 Privacy, cybersecurity, and secure development (SEC) - 72 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-001-01-01	Cybersecurity and privacy: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-001-01-02	Cybersecurity and privacy: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-001-01-03	Cybersecurity and privacy: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-001-02-01	Cybersecurity and privacy: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-001-02-02	Cybersecurity and privacy: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-001-02-03	Cybersecurity and privacy: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-001-03-01	Cybersecurity and privacy: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-001-03-02	Cybersecurity and privacy: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-001-03-03	Cybersecurity and privacy: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-002-01-01	Cybersecurity and privacy: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-002-01-02	Cybersecurity and privacy: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-002-01-03	Cybersecurity and privacy: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-002-02-01	Cybersecurity and privacy: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-002-02-02	Cybersecurity and privacy: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-002-02-03	Cybersecurity and privacy: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-002-03-01	Cybersecurity and privacy: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-002-03-02	Cybersecurity and privacy: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-002-03-03	Cybersecurity and privacy: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-003-01-01	Cybersecurity and privacy: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-003-01-02	Cybersecurity and privacy: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-003-01-03	Cybersecurity and privacy: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-003-02-01	Cybersecurity and privacy: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-003-02-02	Cybersecurity and privacy: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-003-02-03	Cybersecurity and privacy: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-003-03-01	Cybersecurity and privacy: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-003-03-02	Cybersecurity and privacy: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-003-03-03	Cybersecurity and privacy: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-004-01-01	Cybersecurity and privacy: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-004-01-02	Cybersecurity and privacy: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-004-01-03	Cybersecurity and privacy: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-004-02-01	Cybersecurity and privacy: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-004-02-02	Cybersecurity and privacy: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-004-02-03	Cybersecurity and privacy: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-004-03-01	Cybersecurity and privacy: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-004-03-02	Cybersecurity and privacy: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-004-03-03	Cybersecurity and privacy: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-005-01-01	Cybersecurity and privacy: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-005-01-02	Cybersecurity and privacy: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-005-01-03	Cybersecurity and privacy: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-005-02-01	Cybersecurity and privacy: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-005-02-02	Cybersecurity and privacy: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-005-02-03	Cybersecurity and privacy: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-005-03-01	Cybersecurity and privacy: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-005-03-02	Cybersecurity and privacy: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-005-03-03	Cybersecurity and privacy: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-006-01-01	Cybersecurity and privacy: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-006-01-02	Cybersecurity and privacy: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-006-01-03	Cybersecurity and privacy: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-006-02-01	Cybersecurity and privacy: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-006-02-02	Cybersecurity and privacy: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-006-02-03	Cybersecurity and privacy: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-006-03-01	Cybersecurity and privacy: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-006-03-02	Cybersecurity and privacy: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-006-03-03	Cybersecurity and privacy: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-007-01-01	Cybersecurity and privacy: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-007-01-02	Cybersecurity and privacy: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-007-01-03	Cybersecurity and privacy: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-007-02-01	Cybersecurity and privacy: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-007-02-02	Cybersecurity and privacy: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-007-02-03	Cybersecurity and privacy: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-007-03-01	Cybersecurity and privacy: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-007-03-02	Cybersecurity and privacy: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-007-03-03	Cybersecurity and privacy: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-008-01-01	Cybersecurity and privacy: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-008-01-02	Cybersecurity and privacy: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-008-01-03	Cybersecurity and privacy: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-008-02-01	Cybersecurity and privacy: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-SEC-008-02-02	Cybersecurity and privacy: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-008-02-03	Cybersecurity and privacy: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-008-03-01	Cybersecurity and privacy: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-008-03-02	Cybersecurity and privacy: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-SEC-008-03-03	Cybersecurity and privacy: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

8.12 Clinical operations, monitoring, and resilience (OPS) - 63 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-001-01-01	Operations, resilience, and monitoring: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-001-01-02	Operations, resilience, and monitoring: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-001-01-03	Operations, resilience, and monitoring: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-001-02-01	Operations, resilience, and monitoring: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-001-02-02	Operations, resilience, and monitoring: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-001-02-03	Operations, resilience, and monitoring: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-001-03-01	Operations, resilience, and monitoring: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-001-03-02	Operations, resilience, and monitoring: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-001-03-03	Operations, resilience, and monitoring: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-002-01-01	Operations, resilience, and monitoring: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-002-01-02	Operations, resilience, and monitoring: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-002-01-03	Operations, resilience, and monitoring: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-002-02-01	Operations, resilience, and monitoring: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-002-02-02	Operations, resilience, and monitoring: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-002-02-03	Operations, resilience, and monitoring: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-002-03-01	Operations, resilience, and monitoring: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-002-03-02	Operations, resilience, and monitoring: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-002-03-03	Operations, resilience, and monitoring: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-003-01-01	Operations, resilience, and monitoring: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-003-01-02	Operations, resilience, and monitoring: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-003-01-03	Operations, resilience, and monitoring: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-003-02-01	Operations, resilience, and monitoring: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-003-02-02	Operations, resilience, and monitoring: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-003-02-03	Operations, resilience, and monitoring: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-003-03-01	Operations, resilience, and monitoring: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-003-03-02	Operations, resilience, and monitoring: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-003-03-03	Operations, resilience, and monitoring: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-004-01-01	Operations, resilience, and monitoring: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-004-01-02	Operations, resilience, and monitoring: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-004-01-03	Operations, resilience, and monitoring: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-004-02-01	Operations, resilience, and monitoring: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-004-02-02	Operations, resilience, and monitoring: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-004-02-03	Operations, resilience, and monitoring: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-004-03-01	Operations, resilience, and monitoring: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-004-03-02	Operations, resilience, and monitoring: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-004-03-03	Operations, resilience, and monitoring: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-005-01-01	Operations, resilience, and monitoring: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-005-01-02	Operations, resilience, and monitoring: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-005-01-03	Operations, resilience, and monitoring: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-005-02-01	Operations, resilience, and monitoring: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-005-02-02	Operations, resilience, and monitoring: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-005-02-03	Operations, resilience, and monitoring: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-005-03-01	Operations, resilience, and monitoring: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-005-03-02	Operations, re-silience, and monitoring: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-005-03-03	Operations, re-silience, and monitoring: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-006-01-01	Operations, re-silience, and monitoring: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-006-01-02	Operations, re-silience, and monitoring: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-006-01-03	Operations, re-silience, and monitoring: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-006-02-01	Operations, re-silience, and monitoring: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-006-02-02	Operations, re-silience, and monitoring: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-006-02-03	Operations, resilience, and monitoring: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-006-03-01	Operations, resilience, and monitoring: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-006-03-02	Operations, resilience, and monitoring: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-006-03-03	Operations, resilience, and monitoring: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-007-01-01	Operations, resilience, and monitoring: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-007-01-02	Operations, resilience, and monitoring: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-007-01-03	Operations, resilience, and monitoring: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-OPS-007-02-01	Operations, re-silience, and monitoring: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-007-02-02	Operations, re-silience, and monitoring: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-007-02-03	Operations, re-silience, and monitoring: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-007-03-01	Operations, re-silience, and monitoring: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-007-03-02	Operations, re-silience, and monitoring: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
SIR-OPS-007-03-03	Operations, re-silience, and monitoring: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

8.13 Verification, commissioning, release, and change control (VAL) - 81 matrices

Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-001-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-001-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-001-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-001-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-001-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-001-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-001-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-001-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-001-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-002-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-002-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-002-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-002-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-002-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-002-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-002-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-002-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-002-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-003-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-003-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-003-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-003-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-003-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-003-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-003-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-003-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-003-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-004-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-004-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-004-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-004-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-004-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-004-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-004-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-004-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-004-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-005-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-005-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-005-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-005-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-005-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-005-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-005-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-005-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-005-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-006-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-006-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-006-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-006-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-006-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-006-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-006-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-006-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-006-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-007-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-007-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-007-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-007-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-007-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-007-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-007-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-007-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-007-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-008-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-008-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-008-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-008-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-008-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-008-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-008-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-008-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-008-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-009-01-01	Verification, commissioning, and release: create one version-controlled boundary record for each deployed producer-consumer route. Each record shall bind...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-009-01-02	Verification, commissioning, and release: publish an executable RP-SCH contract containing a machine-readable schema or DICOM profile, valid and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-009-01-03	Verification, commissioning, and release: place each boundary record, schema or object profile, example set, dependency and compatibility matrix...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-009-02-01	Verification, commissioning, and release: implement an RP-VAL ingress gate at the first trusted boundary. Before accepting a request...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-009-02-02	Verification, commissioning, and release: execute through an RP-TRX controller that rechecks authorization at commit, enforces the approved deterministic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Entity	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
SIR-VAL-009-02-03	Verification, commissioning, and release: use RP-OBS to record immutable request, decision, configuration and model versions where applicable, input...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-009-03-01	Verification, commissioning, and release: maintain an RP-VVP trace matrix that decomposes every shall-clause and assigned family subprofile into...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-009-03-02	Verification, commissioning, and release: execute and retain producer, consumer, integrated-route, representative-workflow, failure-injection, and commissioned-site evidence across the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
SIR-VAL-009-03-03	Verification, commissioning, and release: use RP-RLS to prevent activation or continued clinical availability when trace coverage is incomplete...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

9 Functional component trade matrices

9.1 Functional component - 41 matrices

Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
C-GOV-01	Human authority and approval: Owns intended use, scope, residual-risk acceptance, evidence policy, clinical authority, and release decisions	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
C-QMS-01	Human authority and approval: Controls documents, changes, deviations, suppliers, incidents, CAPA, approvals, and quality records	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-REQ-01	Human authority and approval: Maintains bidirectional links from ConOps through requirements, hazards, components, tests, results, anomalies, and releases	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-ARCH-01	Interoperability and external TPS: Controls system boundaries, service contracts, data ownership, interface-control documents, and component compatibility	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-IAM-01	Human authority and approval: Provides authentication, role and context authorization, separation of duties, approval identity, and access review...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-SAFE-01	Human authority and approval: Evaluates authorization, context, scope, preconditions, hard stops, approval validity, and export eligibility independently of...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-STATE-01	Human authority and approval: Controls draft, review, approval, QA, transfer, degraded, invalidated, and superseded states with atomic transitions	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-UX-01	Human authority and approval: Presents context, provenance, warnings, previews, comparisons, approvals, and human-controlled actions	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-HFE-01	Evidence and clinical rules: Defines critical tasks, formative and summative validation, role curricula, competency, renewal, and use-safety evidence	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
C-VOICE-01	Natural language and intent: Provides bounded speech capture, transcription, safety-critical token flagging, correction, and confirmed transcript output	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-INTENT-01	Natural language and intent: Converts text or confirmed speech into versioned typed intent and validates units, negation, laterality...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-WF-01	Interoperability and external TPS: Coordinates governed tasks, human gates, clinical services, external TPS interactions, QA routing, cancellation, and...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-EVID-01	Human authority and approval: Stores and retrieves institution-approved, versioned, applicable evidence without unrestricted clinical-mode internet retrieval	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-RULE-01	Human authority and approval: Evaluates rule applicability, units, authority order, conflicts, life-cycle state, and no-applicable-rule outcomes	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-CASE-01	Patient and case identity: Binds patient, course, studies, frames, prescriptions, anatomy, prior treatment, task, mode, and destination into...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-REG-01	Prescription, plan, and dose: Invokes commissioned registration and accumulation methods and retains transforms, visualization, uncertainty, approval, and lineage	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
C-SEG-01	Anatomy, registration, and geometry: Controls contour-model applicability, draft generation, uncertainty, validation, editing, review, approval, and nomenclature	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-PLAN-01	Prescription, plan, and dose: Builds structured planning intent and coordinates commissioned optimizers, dose services, robustness, diversity, and candidate...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-DOSE-01	Prescription, plan, and dose: Provides a controlled adapter to commissioned planning, optimization, robustness, and dose engines without generative...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-REV-01	Prescription, plan, and dose: Presents consistent metrics, tradeoffs, spatial dose, robustness, uncertainty, provenance, comments, and role-specific decisions	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-QA-01	Verification, commissioning, and release: Routes independent calculation, patient-specific QA, transfer QA, technique checks, and physics review while preserving...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-DICOM-01	Human authority and approval: Provides approved DICOM query, retrieve, storage, import, export, and routing for imaging and radiotherapy...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-DICOM-02	Anatomy, registration, and geometry: Validates identities, UIDs, references, frames, geometry, units, required attributes, completeness, and source-to-destination fidelity	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
C-TPS-01	Human authority and approval: Encapsulates each commissioned external TPS, its DICOM/DICOM-RT conformance statement, supported workflows, optional approved API...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-FHIR-01	Human authority and approval: Implements approved profiles or guides for workflow and non-DICOM clinical exchanges where deployment scope...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-TERM-01	Prescription, plan, and dose: Controls codes, structure names, units, dose basis, coordinate systems, mappings, conversions, and rounding rules	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
C-OBJ-01	Human authority and approval: Stores immutable object versions, lifecycle state, ownership, integrity values, approvals, and source linkage	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-PROV-01	Anatomy, registration, and geometry: Traces derived objects to inputs, transformations, software, models, evidence, configurations, users, approvals, and downstream...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-AUD-01	Human authority and approval: Records ordered commands, context, evidence, versions, outputs, warnings, edits, approvals, QA, transfers, failures, overrides...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-CFG-01	Human authority and approval: Locks approved scope, modes, rules, thresholds, services, models, machines, interfaces, operating responses, and release...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
C-MGW-01	Human authority and approval: Resolves approved locked models, validates intended use and input contracts, executes inference, and enforces...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-MREG-01	AI model and inference: Controls model versions, provenance, intended use, artifacts, signatures, configurations, limitations, dependencies, and software bills...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-MVAL-01	AI model and inference: Manages evaluation datasets, subgroup and calibration measures, failure detection, edit burden, thresholds, drift, and...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-SEC-01	Anatomy, registration, and geometry: Provides identity federation, encryption, segmentation, secrets, egress control, artifact verification, detection, and privacy safeguards	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-MON-01	Human authority and approval: Calculates safety, model, evidence, workflow, reliability, subgroup, and security signals and invokes approved actions	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-JOB-01	General control realization: Controls pending, running, complete, failed, canceled, timed-out, partial, commit, cleanup, and reconciliation states	C	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
C-RES-01	Operations, resilience, and monitoring: Provides validated backups, staged restoration, downtime and read-only modes, known-good rollback, and reconciliation	C	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
C-INC-01	Evidence and clinical rules: Supports reporting, triage, containment, patient-impact assessment, investigation, evidence preservation, trends, CAPA, and closure	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-VV-01	Cybersecurity and privacy: Executes inspection, analysis, test, demonstration, HFE, site acceptance, commissioning, regression, shadow, and end-to-end protocols	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-TEST-01	Interoperability and external TPS: Controls representative, boundary, wrong-context, failure, adversarial, recovery, interoperability, and clinical challenge cases with expected...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
C-SBX-01	Human authority and approval: Separates research, training, commissioning, shadow, clinical draft, approved, degraded, and read-only execution and destinations	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

10 Interface component trade matrices

10.1 Interface component - 57 matrices

Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-BND-01	Interoperability and external TPS: Creates the canonical identity for every human, service, file, message, DICOM, DICOMweb, IHE, and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-ICD-01	Human authority and approval: Versions and approves the interface-control record, diagrams, responsibility boundaries, authority limits, normal and degraded...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-CAP-01	Human authority and approval: Resolves the approved producer, consumer, vendor, version, site, machine, modality, technique, SOP Class, role...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-OWN-01	Human authority and approval: Assigns accountable producer, consumer, clinical, systems, data, security, operations, supplier, support, and independent-verification authorities...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-CFG-01	Interoperability and external TPS: Publishes signed effective route, endpoint, trust, schema, terminology, timeout, retry, monitoring, feature, scope, and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-RLS-01	Human authority and approval: Allows activation or continued use only when required approvals, compatibility, risk controls, tests, commissioning...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-CHG-01	Cybersecurity and privacy: Detects contract or endpoint changes, performs impact analysis, defines migration and regression scope, coordinates...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-SCH-01	Interoperability and external TPS: Stores machine-validatable message, API, event, file, DICOM object, DICOMweb resource, and IHE content profiles...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-SEM-01	Prescription, plan, and dose: Controls clinical meaning, terminology, nomenclature, units, dose basis, coordinate systems, rounding, state meaning, authority...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-OBJ-01	Human authority and approval: Defines required identifiers, references, lifecycle fields, approval state, provenance, integrity, status, and family-specific content...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-EXM-01	Operations, resilience, and monitoring: Maintains controlled valid, invalid, boundary, wrong-context, stale-version, private-attribute, partial, and recovery examples linked to...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IC-VER-01	Human authority and approval: Negotiates or rejects contract, profile, model, configuration, endpoint, and dependency versions against the approved...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-GTW-01	Interoperability and external TPS: Terminates the first trusted boundary, authenticates the route, applies allowlists and size or rate...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-CTX-01	Patient and case identity: Validates user, patient, course, study, task, object, plan, frame, operating mode, destination, source, freshness...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-REF-01	Prescription, plan, and dose: Validates identifiers, UIDs, reference graphs, frames, transforms, dimensions, orientation, spacing, structure linkage, dose quantity...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-POL-01	Human authority and approval: Evaluates human and service identity, role, purpose, case access, operating mode, intended-use scope, separation...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-SAF-01	Human authority and approval: Enforces deterministic preconditions, allowlisted transitions, hard stops, draft and approved separation, invalidation, export eligibility...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-INT-01	Evidence and clinical rules: Checks hashes or signatures where applicable, object completeness, replay and staleness, configuration and model...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-QRT-01	General control realization: Isolates malformed, incompatible, orphaned, partial, conflicting, suspicious, or unacknowledged content with reason, ownership, retention...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IC-ORC-01	Human authority and approval: Coordinates producers, validators, human gates, clinical services, external systems, QA, acknowledgment, cancellation, recovery, and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-TXN-01	Human authority and approval: Applies one approved state mutation or an explicit compensation, rechecks authorization at commit, and...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-IDM-01	Operations, resiliency, and monitoring: Assigns request keys and sequence rules, detects replay, duplicate, stale, out-of-order, and conflicting work...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-ACK-01	Human authority and approval: Returns one correlated accepted, rejected, canceled, partial, failed, or degraded outcome and distinguishes transport...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-CMP-01	Human authority and approval: Executes the approved cleanup, compensating action, prior-version restoration, approval invalidation, and reconciliation plan without...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-TMO-01	Operations, resilience, and monitoring: Enforces bounded timeouts, backoff, retry limits, cancellation, late-result handling, dead-letter behavior, and safe user-visible...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IC-AUD-01	Human authority and approval: Records correlated requests, decisions, versions, input and output identities, actors, timing, state changes, outcomes...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-PRV-01	Anatomy, registration, and geometry: Links exchanged and derived objects to sources, transforms, schemas, services, models, configurations, evidence, users...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-MON-01	AI model and inference: Calculates availability, latency, queue, validation, rejection, duplicate, partial, reconciliation, drift, security, and safety signals...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-REC-01	Interoperability and external TPS: Compares producer, consumer, repository, TPS, OIS, and destination identities, content, references, states, acknowledgments, and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-ALT-01	Evidence and clinical rules: Routes safety, clinical, interoperability, security, availability, and supplier events to accountable teams with severity...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-BDR-01	Operations, resilience, and monitoring: Provides validated backup, restore, read-only or degraded operation, known-good configuration recovery, route reactivation, and...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IC-TIM-01	Evidence and clinical rules: Maintains trusted time, clock-health evidence, correlation identifiers, ordering metadata, and explicit handling when timing...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-DCM-01	Human authority and approval: Implements commissioned association negotiation and approved SCU or SCP services, SOP Classes, transfer syntaxes...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-DWP-01	Human authority and approval: Implements approved user-agent or origin-server QIDO-RS, WADO-RS, STOW-RS, or other scoped DICOMweb behavior with...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-RTP-01	Prescription, plan, and dose: Applies site-approved RT object profiles and validates prescription, structure, plan, ion plan, dose, registration...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-DVA-01	Interoperability and external TPS: Performs paired conformance analysis, capability comparison, object validation, private-attribute policy, semantic round-trip comparison, and...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-TPS-01	Human authority and approval: Encapsulates each vendor, product version, site, machine, modality, technique, route, workflow, limitation, acknowledgment, and...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-VAP-01	Human authority and approval: Exposes only approved vendor operations with explicit authentication, authorization, schema, version, timeout, error, audit...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-IHE-01	Human authority and approval: Implements approved IHE-RO actor, transaction, content-module, option, and version behavior and reconciles profile workflow...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-UIB-01	Human authority and approval: Connects the clinical workspace to context, preview, comparison, warning, confirmation, cancellation, annotation, approval, and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-NLG-01	Natural language and intent: Connects typed text and push-to-talk speech to confirmed transcript, typed intent, terminology, units, ambiguity...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-EVG-01	Human authority and approval: Connects the closed approved evidence service and computable-rule resolver to source authority, applicability, conflict...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-IMG-01	Prescription, plan, and dose: Connects case imaging, registration, accumulated dose, segmentation or contours, uncertainty, edit lineage, review, and...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-PLN-01	Prescription, plan, and dose: Connects structured planning intent to commissioned optimization, dose, robustness, machine, technique, candidate, and calculation...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-REV-01	Human authority and approval: Connects candidates to comparison, selection, professional approval, physics review, independent verification, patient-specific QA, transfer...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-MDG-01	Human authority and approval: Connects only approved signed model packages to contract-validated inputs, inference, uncertainty, abstention, draft outputs...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-DAT-01	Data integrity, provenance, and audit: Connects services to immutable versioned objects, terminology, identity, dependency lineage, integrity, audit, retention, lifecycle...	I	3/3/3/3/3/5; 3.20	4/4/5/4/4/3; 4.05	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
IC-IAM-01	Human authority and approval: Federates identity and maps actor or service, role, purpose, case, session, signature, approval meaning...	I	1/3/2/2/2/5; 2.15	4/4/5/4/4/3; 4.05	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
IC-SEC-01	Anatomy, registration, and geometry: Provides mutual end-point trust, approved encryption, network segmentation, certificate and secret lifecycle, rate and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-PTM-01	Human authority and approval: Enforces purpose-bound minimum-necessary content, field and object allowlists, de-identification where approved, destination and residency...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
IC-TRC-01	Human authority and approval: Maintains bidirectional links from ConOps, HLR, HLIR, SIR, family profile, hazard, component, contract, test...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-TST-01	Cybersecurity and privacy: Executes schema, semantic, role, state, security, positive, negative, boundary, wrong-context, stale-version, timeout, duplicate, cancellation...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-ITL-01	Interoperability and external TPS: Provides controlled producer, consumer, simulator, capture, reference-comparison, failure-injection, load, multi-vendor, version, route, and semantic...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-COM-01	Human authority and approval: Binds the actual institution, vendor, product version, endpoint, machine, modality, technique, disease site, workflow...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-HFE-01	Human authority and approval: Evaluates representative users and critical tasks including context confirmation, speech correction, warning, interruption, comparison...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-AIE-01	AI model and inference: Tests the combined model and interface route for input contract, intended-use scope, subgroup behavior...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
IC-RPT-01	Human authority and approval: Produces reproducible signed coverage, actual-result, anomaly, deviation, residual-risk, limitation, approval, commissioning, release, monitoring, and...	I	2/3/2/2/3/5; 2.60	4/4/5/4/4/3; 4.05	5/4/5/5/2/1; 4.00	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

11 Category component trade matrices

11.1 Functional category component - 30 matrices

Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
FC-REQ-01	Human authority and approval: Maintains bidirectional trace from ConOps and source HLRs through categorized requirements, hazards, tests, results...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-ARCH-01	Interoperability and external TPS: Controls functional boundaries, service contracts, data ownership, interface dependencies, and compatible component versions	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-CFG-01	Human authority and approval: Locks approved functional scope, workflows, parameters, models, evidence, machines, interfaces, and release manifests	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-PROV-01	Anatomy, registration, and geometry: Links each input, action, transformation, result, approval, and transfer to the exact actor, object...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-VV-01	Evidence and clinical rules: Executes positive, negative, boundary, recovery, interoperability, and end-to-end tests with retained objective evidence	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-IAM-01	Human authority and approval: Authenticates users and services and enforces role, context, approval identity, and separation of duties	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-SAFE-01	Human authority and approval: Evaluates scope, context, preconditions, hard stops, human authority, approval validity, and export eligibility independently...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
FC-STATE-01	Human authority and approval: Controls draft, review, approval, QA, transfer, invalidated, degraded, and superseded object and workflow states	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-UX-01	Human authority and approval: Presents context, sources, assumptions, warnings, previews, comparisons, edits, approvals, and human-controlled actions	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-VOICE-01	Natural language and intent: Provides bounded speech capture, transcription, critical-token flagging, correction, read-back, and confirmed transcript output	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-INTENT-01	Natural language and intent: Converts text or confirmed speech into inspectable typed intent and validates objects, units, laterality...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-WF-01	Interoperability and external TPS: Coordinates governed clinical tasks, human gates, commissioned services, existing TPS interactions, QA routing, cancellation...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-EVID-01	Human authority and approval: Stores and retrieves institution-approved, versioned, applicable literature and local policies without unrestricted clinical-mode internet...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-RULE-01	Human authority and approval: Evaluates evidence applicability, units, authority, conflicts, lifecycle state, and no-applicable-rule outcomes	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
FC-CASE-01	Patient and case identity: Binds patient, course, studies, frames, prescription, anatomy, prior treatment, task, mode, and destination	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-REG-01	Prescription, plan, and dose: Invokes commissioned registration and accumulation methods and retains transforms, uncertainty, approval, and lineage	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-SEG-01	Anatomy, registration, and geometry: Controls contour-model applicability, draft generation, uncertainty, editing, review, approval, and nomenclature	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-PLAN-01	Prescription, plan, and dose: Builds structured planning intent and coordinates candidate generation, diversity, optimization, dose, and robustness services	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-DOSE-01	Prescription, plan, and dose: Connects only to commissioned optimization, robustness, and dose engines; it does not permit generative...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-REV-01	Prescription, plan, and dose: Presents consistent metrics, spatial dose, trade-offs, robustness, uncertainty, provenance, comments, and role-specific decisions	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-QA-01	Verification, commissioning, and release: Routes independent calculation, patient-specific QA, transfer verification, technique checks, and physics review while preserving...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
FC-DICOM-01	Human authority and approval: Provides approved query, retrieve, store, import, export, validation, and reconciliation for images and radiotherapy...	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-TPS-01	Human authority and approval: Encapsulates each commissioned TPS, its DICOM conformance, supported workflows, optional approved API, limitations, and...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-TERM-01	Prescription, plan, and dose: Controls codes, structure names, units, dose basis, coordinate systems, mappings, conversions, and rounding	C	1/3/2/2/2/5; 2.15	4/4/4/4/4/3; 3.90	5/4/5/5/3/1; 4.15	C; Gate. Independent or redundant control is preferred. Use B only if common-cause, latency, usability, or feasibility analysis demonstrates equal or lower residual risk.
FC-OBJ-01	Human authority and approval: Stores immutable object versions, lifecycle state, ownership, integrity values, approvals, and source links	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-MGW-01	Human authority and approval: Resolves approved locked models, validates intended use and input contracts, executes inference, and enforces...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-MREG-01	AI model and inference: Controls model versions, provenance, intended use, artifacts, signatures, configurations, limitations, and dependencies	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
FC-MVAL-01	Evidence and clinical rules: Manages validation datasets, subgroup and calibration measures, edit burden, thresholds, drift, and regression evidence	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
FC-JOB-01	General control realization: Controls pending, running, complete, failed, canceled, timed-out, partial, commit, cleanup, and reconciliation states	C	3/3/3/3/5; 3.20	4/4/4/4/3; 3.90	4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
FC-AUD-01	Human authority and approval: Records ordered commands, context, evidence, outputs, warnings, edits, approvals, QA, transfers, failures, and reconciliation	C	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

11.2 Non-functional category component - 22 matrices

Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NFC-REQ-01	Human authority and approval: Maintains non-functional requirements, hazards, controls, verification, anomalies, residual-risk decisions, and release trace	C	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-ARCH-01	Cybersecurity and privacy: Defines safety, usability, privacy, security, integrity, observability, and fail-safe contracts at component and interface...	C	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-CFG-01	Human authority and approval: Locks quality-attribute thresholds, safety policies, warning tiers, trust rules, security baselines, and approved exceptions	C	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-MON-01	Human authority and approval: Collects versioned safety, usability, security, integrity, and control-effectiveness measures with approved response thresholds	C	2/3/2/2/3/5; 2.60	4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NFC-VV-01	Evidence and clinical rules: Runs safety, misuse, human-factors, security, resilience, integrity, and negative tests with retained acceptance evidence	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-SAFE-01	Human authority and approval: Implements fail-closed preconditions, authority limits, hard stops, approval invalidation, and safe export rules	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-STATE-01	Human authority and approval: Controls valid states and atomically invalidates affected approvals, QA, and transfer eligibility when dependencies...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-IAM-01	Human authority and approval: Enforces unique identity, current role and context, least privilege, clinical approval identity, and incompatible-role...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-CTX-01	General control realization: Creates immutable action context, validates patient and object identity, and rejects stale, incomplete, or...	C	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
NFC-HFE-01	Human factors and use safety: Controls critical-task analysis, formative and summative evaluation, foreseeable misuse, training, competency, and remediation	C	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
NFC-UX-01	Operations, resilience, and monitoring: Provides persistent context, semantic source labels, accessible multimodal status, confirmations, and usable recovery paths	C	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
NFC-ALERT-01	Interoperability and external TPS: Controls risk tier, presentation, acknowledgment, escalation, suppression, burden measures, and corrective review	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NFC-SEC-01	Anatomy, registration, and geometry: Provides segmented trust boundaries, least-privilege services, secure configuration, endpoint control, detection, and enforcement	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-PRIV-01	Human authority and approval: Classifies protected data, enforces permitted purpose and destination, and limits disclosure to approved minimum-necessary...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-CRY-01	Human authority and approval: Controls approved encryption, keys, secrets, certificates, rotation, revocation, status, and recovery	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-EGRESS-01	Cybersecurity and privacy: Allowlists connectors, tenants, destinations, data classes, and purposes and blocks incompatible protected-data transfer	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-INPUT-01	Human authority and approval: Separates external content from policy and tool authority and sanitizes or rejects embedded authority-changing...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-ART-01	Human authority and approval: Validates manifests, signatures, provenance, integrity, compatibility, approval, and revocation before clinical load or use	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-SDL-01	AI model and inference: Controls threat modeling, secure coding, review, vulnerability management, penetration testing, patching, SBOM, and supplier...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
NFC-AUD-01	Evidence and clinical rules: Retains synchronized, integrity-protected evidence of context, controls, warnings, access, transfer, exceptions, incidents, and response	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-DATA-01	Cybersecurity and privacy: Validates completeness, identity, consistency, version, integrity, and provenance of data used by safety or...	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
NFC-INC-01	Evidence and clinical rules: Coordinates triage, containment, patient-impact assessment, evidence preservation, communication, recovery, CAPA, and closure	C	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

11.3 Operational category component - 22 matrices

Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OC-REQ-01	Human authority and approval: Maintains operational needs, procedures, risks, controls, exercises, results, anomalies, approvals, and release trace	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-GOV-01	Human authority and approval: Owns intended use, scope, clinical authority, residual-risk acceptance, evidence policy, and accountable decisions	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-QMS-01	Human authority and approval: Controls documents, changes, deviations, incidents, CAPA, approvals, records, and quality review	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OC-CFG-01	Human authority and approval: Binds approved scope, modes, controls, thresholds, services, models, interfaces, procedures, and release manifests	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-AUD-01	Human authority and approval: Retains decisions, events, exercises, exceptions, results, approvals, actions, and immutable operational chronology	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-VV-01	Operations, resilience, and monitoring: Executes inspections, exercises, commissioning, recovery, shadow, site-acceptance, and end-to-end operational protocols	O	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OC-SCOPE-01	Human authority and approval: Controls approved institutions, users, populations, sites, modalities, machines, techniques, tasks, and exclusions	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-MODE-01	Human authority and approval: Segregates research, training, commissioning, shadow, clinical-draft, clinical-approved, degraded, and read-only modes and destinations	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-ACCESS-01	Human authority and approval: Administers role matrices, competency dependencies, exceptions, separation of duties, and periodic access review	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-REL-01	Human authority and approval: Confirms approved baseline, residual risk, verification, commissioning, training, readiness, and authorization before clinical activation	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OC-CHG-01	Human authority and approval: Classifies changes, assesses impact, defines regression and migration, approves activation, and preserves known-good rollback	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-MON-01	Evidence and clinical rules: Monitors service health, workflow, model, evidence, interface, safety, quality, and subgroup indicators under controlled...	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-ALERT-01	Interoperability and external TPS: Evaluates thresholds, opens actionable records, routes escalation, records acknowledgment, and tracks disposition	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-INC-01	Evidence and clinical rules: Supports reporting, triage, containment, patient-impact assessment, evidence preservation, investigation, communication, and closure	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-CAPA-01	General control realization: Controls root-cause analysis, assigned actions, due dates, effectiveness checks, trend review, and quality closure	O	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OC-BKP-01	Human authority and approval: Protects approved objects and configurations and validates encrypted backup, retention, restoration authority, and restore...	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-DOWN-01	Operations, resilience, and monitoring: Defines safe access, external work practices, communication, restrictions, reconciliation, and controlled restoration	O	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.

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Component	Decision focus	Set	A: S/U/I/A/L/C; W	B: S/U/I/A/L/C; W	C: S/U/I/A/L/C; W	Preferred / sensitivity
OC-REC-01	Operations, re-silience, and monitoring: Exercises recovery objectives, known-good restoration, deviations, corrective actions, and continued adequacy	O	3/3/3/3/3/5; 3.20	4/4/4/4/4/3; 3.90	4/4/4/4/2/1; 3.40	B; Moderate. The bounded baseline balances control and lifecycle burden. A requires full compliance evidence; choose C if monitoring, incidents, change, or scope raises risk.
OC-COMM-01	Human authority and approval: Controls local configuration, machine and TPS interfaces, reference cases, clinical workflow tests, acceptance, and...	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-TEST-01	Cybersecurity and privacy: Controls representative, boundary, failure, misuse, recovery, interoperability, and clinical cases with expected results	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-ANOM-01	Human authority and approval: Classifies discrepancies, blocks inappropriate acceptance, records investigation, and requires approved disposition and retest	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.
OC-REV-01	Evidence and clinical rules: Reviews field signals, scope, residual risk, controls, evidence, models, changes, and continued authorization at...	O	2/3/2/2/3/5; 2.60	4/4/4/4/4/3; 3.90	5/4/4/5/2/1; 3.85	B + targeted C; High. Use the bounded baseline and add independent validation, reconciliation, or isolation where residual risk, regulation, commissioning, or separation of duties requires it.

12 Decision gate and required trade-study record

Before implementation or release, each material decision shall identify the decision authority, alternatives considered, compliance screen, evidence sources, criterion definitions, weights, scores, uncertainty ranges, sensitivity to weights and assumptions, risk-control effects, cybersecurity and human-factors effects, common-cause and dependency risks, cost and schedule basis, dissent, final selection, conditions, monitoring, and change triggers.

A preferred C disposition does not automatically require a second full system. The team shall identify the minimum independence or diversity needed to reduce the specific hazard without introducing inconsistent state, latency, alert burden, new attack surface, or unmanageable support complexity. A preferred B plus targeted C disposition means baseline componentization with independent safeguards only at the risk-significant decision or data boundary.

13 Structural trade-matrix audit

Entity set	Expected	Unique IDs	Matrices	Missing	Audit
System high-level requirements	111	111	111	0	Pass
Detailed sub-requirements	333	333	333	0	Pass
High-level interface requirements	333	333	333	0	Pass
Sub-interface requirements	999	999	999	0	Pass
Functional components	41	41	41	0	Pass
Interface components	57	57	57	0	Pass
F/NF/O category components	74	74	74	0	Pass
TOTAL	1,948	1,948	1,948	0	Pass

Structural pass establishes complete matrix coverage only. It does not validate alternatives, weights, scores, evidence, uncertainty, sensitivity, feasibility, risk reduction, architecture selection, procurement, or clinical release.

Bottom line. Every unique controlled requirement and component now has a preliminary, traceable trade-off decision matrix that preserves mandatory obligations and makes safety, clinical utility, interoperability, assurance, lifecycle, and delivery burden explicit.